University of Innovation
جامعة الابتكار

Innovation University

Faculty of
Computers and Information Technology (FCIT)

Undergraduates Internal Regulations and Curricula (Credit Hours System)

January 2024

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Part A: Introduction

Faculty of Computers and Information Technology (FCIT) is one of Innovation University faculties which established according to the presidential decree number 634 for year 2021.

Vision, Mission, and Objectives of FISCS

Vision

The Faculty of Computers and Information Technology shall achieve excellence in the fields of information systems, computer science, artificial intelligence, and digital informatics, and that we be creative in developing the learning journey for students who are the focus of the educational process, and that we have leadership in our specialty at the local and regional levels.

Mission

The Faculty of Computers and Information Technology is committed to providing students with basic knowledge and skills and providing an educational environment conducive to innovation in the field of computers and digital informatics with a balance between the ethical, professional, technical, and scientific aspects, and graduating highly competitive professionals in the fields of computer science, information systems, artificial intelligence, digital twins, and to contribute to advancing technological development in all its aspects for our dear country, Egypt.

Objectives

The Faculty of Computers and Information Technology's objectives are:

1. Developing competitiveness through an advanced educational system that suits the rapid growth in the fields of computers and information
2. Developing scientific creativity to keep pace with modern scientific trends in the faculty's fields of specialization
3. Developing innovation and creativity skills
4. Providing a standard environment and a typical climate for the educational process
5. Developing practical creativity to keep pace with the requirements of institutions concerned with specialization.
6. Developing communication and teamwork skills
7. Developing business management and leadership skills
8. Developing students' awareness of ethical standards
9. The ability to self-learn for life
10. The ability of dynamism/agility/resilience/flexibility

Excellence of this Curriculum

This curriculum emphasizes the importance of self-directed learning for the student, making the student the focus of the educational process, and enabling learning through modern and renewable educational technologies, allowing teaching and learning, whether on-site or remotely, with stimulation, excitement, and enjoyment, and implementing new educational strategies such as problem-based learning. And other strategies enabled by technology in the presence of a comprehensive educational infrastructure that includes digital learning coding and other resources. In addition to this, the transition from curricula directed towards a single science to integrated curricula. The following concepts are the basis for designing this regulation:

1. All students will have engaging and empowering learning experiences in both formal and informal settings that prepare them to be active, creative, knowledgeable, and ethical participants in our society. Accordingly, we need to
    - Switch from education to learning.
    - Recognize the student as the core of the teaching process.
    - Provide students with best environment to succeed in their studies.
    - Focus on practical applications in the field.
2. Provide multiple means of representation so that students can approach information in more than one way.
3. Provide multiple means of expression so that all students can demonstrate and express what they know.
4. Provide multiple means of engagement to stimulate idlers in and motivation for learning.
5. One Campus, one curriculum, one standard quality of education throughout the campus.

Article (1): Faculty Departments

The courses at Faculty of Computers and Information Technology (FCIT)- Innovation University are offered by Four departments (see Table 1).

Table 1 List of Departments at FCIT

#	Field	Department	Acronym
1	**Computer Science**	Computer Science Department	**CS**
2	**Information Systems**	Information Systems Department	**IS**

This is in addition to a department for basic sciences, humanities, and other like courses taught from outside Faculty Programs, the acronym is **BS**.

The courses at the Faculty of Computers and Information Technology FCIT, Innovation University, are offered by two departments. The Faculty Department is responsible for scientific research, and teaching courses to all programs, which need courses in the specialization of the department and holding the department code. The department is responsible for the scientific coding of the course and the nomination of instructors to each course, either from the department, from another department, or from outside the faculty. The department is responsible for the continuous development of teaching methodologies and scientific coding of the courses.

Faculty of Computers and Information Technology (FCIT) may establish other departments in accordance with academic and research developments at the local / regional / international levels and shall be subject to the provisions of the University Regulation Law. The Faculty Council assigns one or more departments to supervise general courses, basic sciences, humanities, and other like courses taught from outside the faculty.

Article (2): Offered Programs

Faculty of Computers and Information Technology (FCIT) - Innovation University offers a variety of programs. Each program is administrated by a Program Steering Committee. The programs are divided into specialized and interdisciplinary programs. They are carefully selected to satisfy the needs of the national industry, as well as, the needs of the regional industry, which recruits many graduates from Egyptian universities.

Innovation University awards, based on the request of the Council of the Faculty of Computers and Information Technology (FCIT), the Bachelor of Computers and Information Degree in the following (Provided that there are a number of foundation hours that can be completed over two semesters that are shared by all programs) :

Table 2 List of Undergraduate Programs offered by FCIT.

Computers & Information Technology	1	Computer Science Program	CS
	2	Information Systems Program	IS
	3	Artificial Intelligence Program	AI
	4	Digital Informatics Program	DI

1. Bachelor of Computers and Information in Computer Science

The department AIs to prepare specialized and highly qualified human cadres in computer science and its various fields such as computer programming, concepts of computer languages and their translations - data structures, organization and processing of files - analysis and design of algorithms - computer operating systems - computer structure and organization - software engineering - foundations and applications of artificial intelligence - systems Smart - natural language processing - multi-agent systems - genetic algorithms - knowledge base systems - parallel processing and distributed systems - computer education - human-computer communication methods - cloud computing - software security - applied data science - software testing methods - knowledge engineering - systems

2. Bachelor of Computers and Information in Information Systems

The department AIs to prepare specialized and highly qualified human cadres in Information systems and its various fields such as Such as analysis and design of information systems - information systems development methodologies - information systems architectures - information storage and retrieval systems - database systems - management information systems - geographic information systems - multimedia information systems – distributed information systems - large data systems - data science - systems Intelligent information - prospecting and tracking of data sources - data warehouses - integrated information systems - methodologies for developing information systems - quality assurance of software and information systems - analysis and exploration in social networks - applications of information systems in various fields - e-commerce - internet information systems - big data systems Scale - Manage and prospect in work processes.

3. Bachelor of Computers and Information in Artificial Intelligence

The department AIs to prepare specialized and highly qualified human cadres in artificial intelligence and its various fields such as the principles of artificial intelligence - machine learning, pattern recognition - robotics - data mining - developing smart interactive applications - understanding natural languages - cognitive computing.

4. Bachelor of Computers and Information in Digital Informatics

The Department of Digital Informatics AIs to prepare specialized and highly competent human cadres in digital informatics and its various fields such as: Fundamental of Artificial Intelligence, Data Visualization and Analytics, Intelligence Agents and Multi Agents Systems, Neurosciences and Connectionist Networks, Computer and Cyber Security, Digital Twins, Speech and Audio Signal Processing, Statistical Artificial Intelligence, Cyber-Physical Systems, AI and e-Health, Internet and Communication Technologies, Multimedia Communications and Networking

The Faculty of Computers and Information Technology may establish other specializations or programs in accordance with the provisions of the University Regulation Law on the recommendation of the relevant department councils and with the approval of the Faculty and the University Councils. It is also permissible not to accept new students in some disciplines based on the decision of the Faculty Council in the case of the lack of appropriatenumbers of students or lack of facilities and possibilities of offering.

Degrees are awarded to students who have completed graduation requirements in three graduation roles at the end of each semester (January - June - September).

Article (3): Agreements with other Universities

The university seeks be partnered with other universities, either by offering double degrees,or through cooperating in several teaching activities.

Article (4): Program Development Committee

The Program Development Committee is assigned by the Faculty Council.The Program Development Committee is responsible for the following:

1. All administration aspects concerned with the program students.
2. The academic advising for the program students.
3. The study plan for each student until completing the program requirements.
4. Administration of the course equivalence for students transferred from other programs or from other faculties. The decision of the course equivalence is done by the concerned department.
5. The educational process in cooperation with the Continuous Quality Assurance Unit (QAU) inside the faculty and reviewing the reports on the instructors.
6. Planning course offering in the beginning of every semester.

Article (5): Education and Student Affairs Committee

The Education and Students Affairs Committee is responsible for studying all issues related to students, such as:

1. Individual students' requests, and excuses for study periods or exams.
2. Individual students' complaints.
3. Students' admission cases and rules.
4. Students' transfer requests from other universities.
5. Approving students' requests for idler-program transfers.
6. Students' affairs processes and procedures.
7. Promoting students' activities.
8. Coordinating students' field training.

All recommendations of the Education and Students' Affairs Committee are presented to the Council of the Faculty for final approval.

Part B: Admission Regulations

Article (6): Placement Tests to newly admitted students.

The study at the Faculty of Computers and Information Technology (FCIT) requires minimum level of English Language and Mathematics. The Faculty Council can organize a placement test for the students enrolled in the Faculty in English and Mathematics. The Faculty Council can ask the students who fail these tests to take none-credit courses to have equal opportunity with other students.

Placement tests in English and Mathematics will be organized for all accepted students. The Education and Student Affairs Committee can study any changes in these rules according to the study needs.

Article (7): Enrollment Requirements and Scholarship System

1. Faculty of Computers and Information Technology (FCIT)- Innovation University is a private Faculty. It offers higher education in specialized programs for defined tuition fees decided by the Faculty Council every year. The students who benefit from this education are those who have completed the Egyptian High School Certificate (Thanaweya Amma certificate- Mathematics or Science Streams) or equivalent (e.g., IGCSE, American Diploma, certificates from other countries approved by the Supreme Council of Private Universities).
2. Science stream students must pass the course corresponding to Math 2 for high school mathematics students, and the time specified for holding the exam is set by University and the exam is held inside the University.
3. All programs in this Bylaw are offered with the Credit-Hour System.
4. If the student fails to achieve a minimum Semester GPA of 2.0 for 4 consecutive main semesters, he/she can be exceptionally allowed to register courses for 2 more semesterswith a maximum 12 credit hours for each semester to improve his/her GPA
5. If the student failed in any course, he/she could register the same for a maximum of four times.
6. Students can only register courses in the main semesters. However, they can register courses in the summer semester with a maximum 9 credit hours.
7. Any additional conditions for admission to the faculty may be requested by the Faculty Council and approved by the University Council and announced at the time of admission.

Article (8): Tuition Fees

1. Tuition fees, set per Credit Hour, are specified yearly and per each semester by Innovation University Council.
2. The tuition fees are paid every semester (the first and second main semesters) based on the number of credit hours registered by the student, with a minimum of the correspondence of educational service fees of 12 CH each semester, unless the number of credit hours remaining for the fulfillment of the degree is less than that, in which case the student should pay the actual number of registered credit hours.
3. Course registration is not final until the student pays the tuition fees for the semester.

Article (9): Student Transfers from other Universities

Students enrolled in a Faculty of Computers & Information (or similar) at another university are allowed to be transferred to the Faculty of Computers and Information Technology (FCIT) - Innovation University, if they fulfill the following conditions:

1. The student could transfer till the third level (Level 3) only from another equivalent level. It is not allowed to transfer students from higher levels to Level 3.
2. It is not allowed to transfer students who are dismissed from their faculty because of exceeding the maximum number of academic probation or failures.
3. Transfer requests are processed by the Student Affairs Department at Innovation University.
4. The Education and Student Affairs Committee can study any changes in these rules according to the study needs.

Part C: Study Regulation Based on Credit Hour System

The articles in this part regulate the course teaching, learning and assessment throughout the programs. These articles are based on the terms of Reference for the design of Credit- Hour Programs for undergraduate level published by the Computers Sector Committee, Supreme Council of Universities on 26 December 2021.

The faculty also includes its study curricula electronically in the distance education system, as well as electronically holding exams to suit the nature of its study whenever its infrastructure

and technological capabilities enables it to (**The Ministerial Decree No. 1200 of 2020**, issued on June 15, 2020).

Article (10): Programs' System and Student Evaluation

1. English is the official language for teaching and examinations. University requirements courses can be taught and examined in Arabic language.
2. The Programs follow the Credit-Hour (CH) system with a total number of 136 credit hours required to obtain the bachelor degree. Senior year students are required to prepare a project over the span of two semesters. The project topic is decided by the different specialized academic departments, and the faculty council decides based on the recommendation of the different academic departments extra time after the semester two written exams for project preparation. The faculty council will appoint the project evaluation committees.
3. The academic year is composed of two semesters, Fall and Spring semesters. The faculty can offer a Summer semester, which will be optional for both the students and the faculty.
4. Each lecture hour is considered one credit hour, and each laboratory or tutorial hour is considered a half-credit hour. Students are not permitted to register for more than 18 credit hours or less than 12 credit hours per semester. In special circumstances, and based on the regulations established by the faculty, recommendation by the academic advisor, and the approval of the Dean, a student can be eligible to register for three additional credit hours beyond the maximum limit if his Cumulative Grade Point Average (CGPA) is not less than 3.5 during the graduating semester, or a CGPA of 2 during the graduating semester.
5. Each academic instructor is responsible for managing the academics of the subject he is teaching based on the specifications prepared by the academic department and approved by the faculty council.
6. Each department prepares complete courses specifications according to the program specification and is approved by the Faculty Council and becomes obligatory for all faculty staff who teaches these courses.
7. Faculty staff/member is also obliged to submit a report on the course he/she has taught to the head of the department at the end of each semester. The head of the department shall submit a report to the vice dean of education and students' affairs on completing all courses reports.
8. The marks of each course are distributed as percentages of the total mark, divided for practical courses into Course Activities (20%), lab activities (20%), Mid-Term Exam (20%), and Final Exam (40%). For theoretical courses, marks are divided into Course Activities (10%), Mid-Term Exam (10%), and Final Exam (80%).
9. The student must attend at least 75% of all course contact hours to be allowed to attend the course final examination.

10. For the student to pass a course, the minimum mark that must be earned in the final exam is 30% of the total exam marks, otherwise the student will fail the course irrespective of the total marks he/she earned in the course, and he/she will get an (F) grade in this course. This provision does not apply to the courses with no final exam.
11. The student fails the course if he/she obtains an (F) grade (less than 50% of the course marks) or did not attend the final examination without submitting a prior excuse that is accepted by the Education and Student Affairs Committee and approved by the Council of the Faculty or was not allowed to attend the final examination because of exceeding the absence percentage or cheating … etc.
12. None-Credit courses are marked as Pass or Fail. The student gets a grade but does not contribute to the cumulative GPA. To pass the course, the student should get at least 50% of the course total marks. Passing all None-Credit courses in the program is a must for graduation.
13. Online learning is allowed in case of Pandemics such as COVID-19, or for any other reason that prohibits students from physical attending the teaching activities.

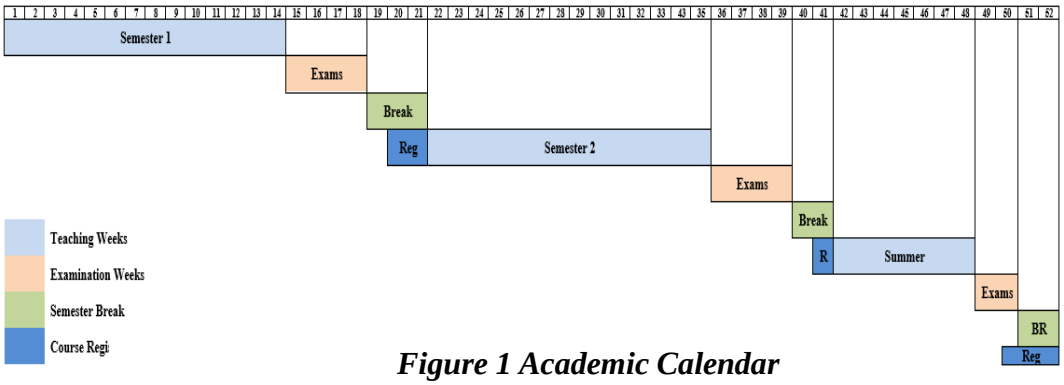

Figure 1 Academic Calendar

Article (11): Study Levels

When the student completes at least 25% of the Program requirements (34 Credit Hours), he/she will be transferred from one level to the next (Level 1 to Level 4). Table 3 shows the student status based on the completed number of achieved CreditHours.

Table 3 Study Levels and relation to the achieved Credit Hours

Study Level	Student Status	Achieved Credit Hours
1	Freshman	Not less than 34 CH
2	Sophomore	Not less than 68 CH
3	Junior	Not less than 104 CH

4	Senior	More than 104 CH

Article (12): Academic Semesters and Course Registration

1. The academic year comprises two main semesters, and one "optional" summer semester:
   - **First semester (Fall):** Begins according to the instructions of private universities counciland lasts for 16 weeks including examinations. Course registration takes place within 3 weeks before the starting day of the semester.
   - **Second semester (Spring):** Begins according to the instructions of private universities council and lasts for 16 weeks including examinations. Course registration takes placewithin 3 weeks before the starting day of the semester.
   - **Summer semester:** Begins late June or early July and lasts for 7 weeks followed by 2 weeks of examination. Course registration takes place within one week before the starting day of the semester.
2. Registration is not final until the full tuition fees of the semester are paid.
3. New students' enrolment in the programs takes place all year long, after fulfilling all the programs requirements and paying the enrolment fees, per the student status.
4. Registration in the Summer semester is optional.

Article (13): Program Study Duration

- The minimum allowed study duration is seven main semesters.
- The maximum allowed study duration is sixteen main semesters (eight years), which doesnot include frozen semesters for reasons accepted by the Faculty Council, after which thestudent is dismissed from the program.

Article (14): Terms of Course Registration

1. The student may register courses in the main semesters with a maximum total Credit Hours according to the following rules (after approval of the Academic Advisor):
   a. For Fall and Spring semesters:
   i. The minimum number of registered credit hours per semester is 9 credit hours, and this requirement can be waived if the remaining credit hours required to graduate is les than 9 credit hours.
   ii. Maximum numbered of registered credit hours:

1. 18 credit hours for newly admitted students (who are not transferring from other equivalent faculties) during their first academic semester in the faculty.
2. 21 credit hours for students that have a CGPA at the beginning of the semester greater than or equal to 3, or for students who will be graduating during that semester.
3. 18 credit hours for students that have a CGPA at the beginning of the semester greater than 2 but less than 3.
4. 15 credit hours for students that have a CGPA at the beginning of the semester greater than 1 but less than 2.
5. 12 credit hours for students that have a CGPA at the beginning of the semester less than 1.
6. In addition to the above credit hours limit requirements, students are allowed to register for one extra course in which that have an incomplete grade.

b. For the Summer semester:

i. The Summer semester is optional to the students.

ii. The maximum allowed credit hours for registration is 9 credit hours.

2. The student may register one additional course to the above limits if this will lead to his/her graduation after the approval of the academic advisor and if this course is offered in his/her program. For interdisciplinary programs, the course will be offered even if it is not normally offered in this academic semester.
3. To enroll in a course with prerequisites, the conditions of success in these previous prerequisite courses must be met or based on the faculty council decision. The prerequisite course is offered in a semester / level lower than the semester / level of the course to be enrolled.

Article (15): Degree Awarding Requirements

- To obtain the Bachelor of Computers and Information Degree, the student must successfully complete the required Credit Hours in one of the programs according to the requirements stipulated in Part D, with a GPA at graduation of at least 2.0.
- The student must pass all zero-credit courses in his/her Program.
- A graduation project is an essential part of all the programs requirements for graduation. The graduation project must be completed over two successive semesters, as per the program requirement, and the student will not graduate unless he/she fulfills the project pass requirements. The student must earn at least 103 Credit Hours Level (including successful passing all of program's mandatory courses of Level Three required for the project) to register for the graduation project. The project is divided along two successive semesters, and the student must register them in their order.
- The student must perform Field Training for Four weeks during his/her study duration.

Article (16): Field Training

1. The student must perform Field Training for Four weeks in an industrial or service facility related to the student's program and must be under the full supervision of the faculty. Itis also possible to perform the training inside the faculty in a similar environment.
2. The training follow-up will be handled by the academic advisor assigned by the Program Steering Committee.
3. Identifying a company official contact person.
4. The student must submit a technical report to his/her academic advisor at the end of the training period.
5. The company should submit a student's training evaluation form to the academic advisorat the end of the training period.
6. The training is divided into periods of 2 weeks at the end of the second and third levels.
7. Training for a period of 6 weeks is allowed, for only one time during the study duration,at the end of third level only.
8. The Field Training is evaluated on Pass / Fail basis and does not count in the cumulative GPA calculation. Student should pass the Field Training to be graduated.
9. The student should pay the supervision fees for the Field Training at a rate of 2 Credit Hours, if applicable, each academic year during which he is performing Field Training.

Article (17): Adding and Dropping a Course

1. The student may add courses according to “course registration rules”, published by Faculty Council each year. Usually, it is during the third week of the main semesters, or the first week of the summer semester.
2. Adding or dropping course(s) should not violate the minimum and maximum number of Credit Hours registered per semester.

Article (18): Withdrawal from a Course

1. The student may withdraw from any course within the first seven weeks of the main semesters or the first Four weeks for the summer semester.
2. The student does not fail the withdrawn course, provided that the withdrawal applicationand approval are finalized within the time limit.

3. The student gets a (W) grade for the withdrawn course and is allowed to register that course (full attendance and performing all activities including examinations) in a following semester.
4. For elective courses, the student is allowed to change it in a following semester if he/she fails to pass it or withdraws from it. This is subject to the approval of the academic advisor and the requirements of his program.
5. The tuition fees will not be refunded for withdrawn courses. The next time the student registers this course, he will have to pay its fees in full. The student, who withdraws froma whole semester without registering any course must pay the minimum tuition fees which is equivalent to 12 CH.

Article (19): Incomplete Course

1. If the student does not attend the final exam for the course in a semester with an excuse accepted by the Student Affairs Committee and approved by the Faculty Council, the course is considered incomplete, and the student receives a grade of (Incomplete) provided that he obtains at least 60% of the grades for the semester work.
2. The student will be recorded for the semester work grade he obtained at the time of studying the course in which he excused himself from taking the final exam. If the student fails to attend the final exam for the incomplete course on the next available date, the student will receive a grade of (Fail) for that course.
3. The grade (Incomplete) is not calculated in the student's cumulative average.
4. On the next available examination date for the incomplete course, the student takes the examination, after paying the re-examination fee, if possible. The marks for this final exam are added to the semester marks to calculate the total grade for this course.

Article (20): Academic Suspension

1. The student can suspend his enrollment (complete withdrawal) from the academic semester based on the regulations established by the faculty and the University.
2. The student will be considered inactive if he did not register for an academic semester, or if he withdraws without an excuse or an approved excuse by the faculty.
3. A student can suspend his academic studies for four consecutive semesters or a maximum of six non-consecutive semesters. If a student exceeds the maximum number of allowed academic suspension semesters with about an accepted excuse by the faculty and the University, he will be dismissed from the University.
4. A student can request suspending his enrollment at the faculty based on the regulations and requirements established by the University.

Article (21): Course Grades

The GPA of each course is calculated based on the marks a student collects during his/her study of this course (Student Activities – Mid Term Exam – Practical Exam (if applicable) – Final Exam). The following table (Table 4) shows how to calculate the GPA based on the collected marks. The student must get a minimum Grade D to pass the course and be considered in the calculation of the Cumulative GPA.

Table 4 Course grades and equivalent GPA

Marks % Collected	Grade	Points
More than 96%	A+	4.0
92% to less than 96%	A	3.7
88% to less than 92%	A-	3.4
84% to less than 88%	B+	3.2
80% to less than 84%	B	3.0
76% to less than 80%	B-	2.8
72% to less than 76%	C+	2.6
68% to less than 72%	C	2.4
64% to less than 68%	C-	2.2
60% to less than 64%	D+	2.0
55% to less than 60%	D	1.5
50% to less than 55%	D-	1.0
less than 50%	F	0
Absence from attending the final exam without an excuse accepted by the faculty Council	F	0
Continuing course for the next semester	PR	No points and the number of academic hours is not calculated within the cumulative averageuntil after completing the course study, whether with success or failure
Incomplete course	I	
Withdrawing from a course	W	

For other courses where the student is registered as a listener (audit), or is only required topass (zero credit courses), are not included in the cumulative GPA, the course grades will be as Tables 5, 6.

Table 5 Grades of zero credit courses

Grade	Explanation
PR	Continuing course for the next semester
F	Fail
W	Withdrawn
I	Incomplete
F	Absence from attending the final exam without an excuse accepted by the Faculty Council

Table 6 General grade (for each semester - upon graduation)

CGPA	Rank
Less than 1.0	Very Weak
1.0 – Less than 2.0	Weak
2.0 – Less than 2.5	Sufficient
2.5 – Less than 3.0	Good
3.0 – Less than 3.5	Very Good
More than 3.5	Excellent

Article (22): Repeating Courses

1. If the student fails a course (gets F grade), he/she should repeat the course:

   a. If the student fails a course, he must repeat it and attend the exams. If he passes the course on his second attempt, he will be awarded the full mark he received but capped at 83 (the highest mark in grade B).

   b. The CGPA will be calculated based on the student's final grade, and each module will be included only once in the CGPA calculation.

   c. The students ' transcript will include all attempts (the fail and pass).

   d. The tuition fees for repeating a course is equivalent to the tuition required for the first attempt.

2. If the student repeats a course he passed with lower than 2 points to improve a CGPA lower than 2 and to avoid dismissal:

   a. If a student has a CGPA at the beginning of the semester that is less than 2 (under academic observation), then the student must improve his CGPA.

   b. A student under academic observation (due to CGPA lower than 2) can repeat any module (in which his previous grade was lower than 2 points) to improve his CGPA to avoid being dismissed, and must attend the course and exam, and he will be awarded the highest mark from both attempts but capped at 83 (the highest mark in grade B).

   c. The course being repeated has to be from the level the student is currently registered in, or one level lower.

d. There is no limit on the number of courses the student (under academic observation due to CGPA being lower than 2) can repeat (given that his grade in the course was lower than 2 points) to improve his CGPA to 2.

e. Only the highest mark in all attempts will be used in the CGPA calculation.

f. All attempts and awarded grades will be include in the transcript.

g. The tuition fees for repeating a course is equivalent to the tuition required for the first attempt.

3. If the student repeats a course he already passed with more than 2 points to improve his CGPA:

a. A student can repeat a course if his CGPA is greater than or equal to 2 only if he has not satisfied the requirements for graduation.

b. A student can repeat a course (if he already passed it with 2 points or more) to improve his CGPA by attending the course and exam, and he will be awarded the highest grade from all attempts.

c. The maximum number of credit hours a student can repeat from the modules that he has already passed with 2 points or more is nine credit hours whether the nine credit hours are used in the same or different courses.

d. The course being repeated has to be from the level the student is currently registered in, or one level lower.

e. Only the highest mark in all attempts will be used in the CGPA calculation.

f. All attempts and awarded grades will be include in the transcript.

Article (23): Study Dismissal and Academic Probation

1. A student gets an academic probation if his/her cumulative GPA at any main semester is less than 2.0 (excluding his first semester in the faculty).
2. In this case, the max allowed hours to be registered during his/her academic probation is12 CH. **Student and his/her guardians must be formerly informed by his academic probation**.
3. A student will be dismissed from the Faculty of Computers and Information Technology, Innovation University, if he/she gets cumulative GPA less than 2.0 in four consecutive semesters excluding summer semesters. If the student's cumulative GPA exceeds 2.0 in any semester, including summer semester, then the number of consecutive academic probation is reset.
4. The student is dismissed if his CGPA is less than 2.0, and is on academic probation for six consecutive or non-consecutive quarters.
5. The student who is exposed to study dismissal due to his/her inability to raise his/her cumulative GPA to at least to 2.0 will be offered an additional and final chance to register in 2 consecutive main semesters and a Summer semester to raise his/her cumulative GPAto at least to 2.0 and achieves the graduation requirements, provided that he/she has successfully completed at least 80% of the total number of credit hours required for graduation and there is a chance for the student to raise his/her cumulative GPA to at least to 2.0.
6. The student will be dismissed from the Faculty of Computers and Information Technology (FCIT) if he/she fails to achieve the graduation requirements during the maximum study duration, which is eight years.

Article (24): Calculation of the Cumulative Grade Point Average (CGPA)

1. Course points achieved by the student are calculated as the number of Credit Hours of this course multiplied by the course grade points according to Table 4.
2. In any semester, the total points earned by the student are calculated as equal to the sumof the courses points the student earned in this semester.
3. The Cumulative GPA at the end of any semester is calculated as the total points achieved by the student in all courses studied divided by the total number of Credit Hours of these courses, taking into consideration the rules relevant to the repeated and improved courses.

$$Cumulative\ GPA = \frac{\sum_{Courses} Grade\ Points * Credit\ Hours}{\sum_{Courses} Credit\ Hours}$$

With approximate to two decimal digits

4. The Semester GPA is calculated as the total points achieved by the student in his/her courses of this semester divided by the total number of Credit Hours of these courses.
5. The Graduation Cumulative GPA is the Cumulative GPA at Graduation, after fulfilling all the graduation requirements. The student cannot get the degree unless he achieves at least a cumulative GPA of 2.0.
6. The ranking of the graduate is determined based on the Graduation Cumulative GPA. In case of equal Graduation Cumulative GPA between two or more students, the ranking will be based on their total accumulative marks, taking into consideration the rules relevant to the repeated and improved courses.
7. The student's Transcript should include all registered courses during the study duration, including those he/she failed, withdrew from, or improved.

Article (25): Declaration of Honor

- For a student to achieve the declaration of honor, he has to fulfill the following conditions:
  - Maintain a cumulative GPA of 3 throughout his study at the Program.
  - Does not fail any course throughout his study at the Program.
  - Did not get any penalty throughout his study at the Faculty.
  - Finishes his program within max of 8 semesters.

Article (26): Academic Advisor

- Every student is assigned an Academic Advisor (his opinion is a consultancy), who follows-up the student academic progress and assists him/her in selecting the courses each semester.
- There can be more than one Academic Advisor in the Program based on the number of students enrolled in the Program.
- The Program Academic Advisor is responsible for helping to:
  - Choose his/her academic path and helps him/her to select courses each semester.

- Choose the field of training.
- Choose the graduation project.

- The Academic Advisor may ask the student to repeat courses which he/she has already passed or ask him/her to register in additional courses to raise his/her cumulative GPA to that required for graduation.
- The academic advisor remains with the student from the time he joins the faculty until he graduates from it.

Article (27): Student Transfer from Another Program

- It is possible to transfer students from another program (either inside or outside the faculty) to any of the programs offered in this bylaw according to the admission regulations in Part B.

Article (28): Student Grievances

- The student can submit an appeal to review his/her course marks within a week from grades announcement, and after paying the required fees in accordance with the faculty regulations regarding this issue.
- In case of general complaint from a course result, the concerned committee reviews the student's marks and gives a decision regarding the marks of this course.

Article (29): Academic Plan

- The minimum number of credit hours required for obtaining the Bachelor of Computers and Information Degree(B.Sc.) is **136 credit** hours in no less than seven semesters and maximum of eight years.
- A final written examination is held for each course at the end of the semester. A student must obtain at least 30% of the final written exam to pass the course.
- Only Student who transferred to level three can select and register his program's elective courses (subject to Academic Advisor Approval and the requirements of his/her program).
- A student may study one elective course from another Program's Elective courses (subject to Academic Advisor Approval and the requirements of his program).
- New, state-of-the-art, courses can be selected to the elective pool of each program upon the approval of the Department/Faculty councils.

Table 7: Indicative curricula codeines by subject area according to NARS 2010

	Subject Area	Credits	Program(s) %	NARS Tolerance %
A	Humanities, ethical and Social Science (university req.)	14	10.3%	8-10
B	Mathematics & Basic Science	24	17.6%	16-18
C	Basic Computing Sciences (institution req.)	39	28.7%	26-28
D	Applied Computing Sciences (specialisation)	42	30.9%	28-30
E	Training (no. of hours = 5)	Zero	3.7%	3-5
F	Graduation Project	8	5.9%	3-5
G	Optional (Institution character-identifying subjects)	9	6.6%	4-16
	Total	136	**100 %**	**100 %**

The academic standards invoked in programs specifications are driven based on the the academic standards established by the National Authority of Quality Assurance and Accreditation of Education in 2009.

Article (30): Structure of the Program(s)

The faculty applies the Credit Hour System which is based on the credit units.

Students should complete **52** modules (plus graduation project and training), total of **140** credit hours, summarized as follows:

8 modules (University Requirements)	(14 credit hours)
8 modules (Mathematics, Basic Sciences,	(24 credit hours)
13 modules Basic Computing Sciences (institution req.)	(39 credit hours)
10 modules Applied Computing Sciences (specialization)	(30 credit hours)
4 modules (Electives)	(12 credit hours)
Project	(8 credit hours)
1 Training	Zero credit
3 modules Optional (Institution character-identifying subjects)	(9 credit hours)

Article (30): Course Codes

The undergraduate courses are coded according to the following scheme.

AA	N1	N2

- **AA**: University/Faculty/Department/Program code offering the course:
- **UR**: University requirement
- **BS**: Basic science
- **CF**: Faculty requirements
- **CS:** Computer science -
- **IS**: Information systems
- **AI:** Artificial Intelligence
- **DI**: Digital Informatics

N1: Class level (1 - 4) in which the course is typically offered.

N2: Sequence (01 - 99) of the courses among its specialized order within the semester.

Article (31): University Requirements – Liberal Arts Courses

The study of liberal arts is one of the main distinct features of Innovation University undergraduate programs that make such programs different from all other programs offered by other universities.

The focus of Innovation University liberal arts requirements is on:

- Creativity and decision making by allowing freedom of thinking.
- Logical and critical thinking.
- Multidisciplinary and synthesizing.
- Promotion of diversity by learning various ways of explaining ideas and phenomena.
- Soft skills including leadership qualities.

The pedagogy used includes:

- Students cindered by maintaining small size classes.
- Active learning through group work, discussions, projects, and problem solving.
- Allowing students to indirect with others beyond their major.

These requirements are specified to help building the knowledge and skills needed for a modern society.

The AIs of the liberal arts courses are to help individuals for:

1- Preparing domestic citizenship and free-thinking personnel.
2- Character building
3- Fostering ambition to use their specialized knowledge in the real world with transferable skills.
4- Exposing to broad fields such as humanities, social sciences, and natural sciences to inculcate in multidisciplinary thinking.
5- Stimulating and satisfying the intellectual curiosity.
6- Developing the ability to identify creative solutions to problems and take action in the future society.

The weight of University Requirements (UR) courses is **14** credits.

Through these courses, students will cover the bases of arts and humanities, social sciences, natural sciences as well as attaining core skills related to critical and creative thinking and technical writing.

University Requirements (Liberal Arts) Core Courses

Each student must take the following **8** core (mandatory) courses. These courses are classified as follows:

A. Arts and Humanities	**Prerequisites**	**Cr Hrs.**
1. UR111 Human Rights	none	1
B. Social Sciences	**Prerequisites**	**Cr Hrs.**
2. UR113 Entrepreneurship and Innovation	none	1
C. Natural Sciences	**Prerequisites**	**Cr Hrs.**
3. UR222 Natural Resources and Sustainability	none	1
D. Key Skills	**Prerequisites**	**Cr Hrs.**
4. UR112 Introduction to Computers	none	3
5. UR122 Marketing & Digital Strategy	none	2
6. UR221 Creative & Scientific Thinking	none	2
7. UR223 Research Methods & Ethics	none	2
8. UR211 Communication Skills	none	2
Total		**14**

Article (32): Faculty Requirements – Core Basic Sciences Courses

Basic sciences courses are equivalent to 24 credit hours, including courses in Mathematics, English, Physics, Electronics, Management, Organizational Behavior, Feasibility Studies, Economics, and Business to build strong basic science background of computing and informatics for students.

Course Code	Course Title	Prerequisites	Credits
BS113	English (1)	none	3
BS114	Physics & Electronics	none	3
BS116	Mathematics (1): Calculus for Computing Science	none	3
BS124	English (2)	BS113	3
BS126	Mathematics (2): Linear Algebra & Applications	BS116	3
BS213	Project Management	none	3
BS214	Probability & Statistics	BS126	3
BS212	Mathematics (3)	BS126	3
Total			**24**

Article (33): Faculty Requirements – Core Computing Courses

Course Code	Course Title	Prerequisites	Credits
CF117	Introduction to Programming	none	3
CF123	Discrete Mathematics & Graph Theory	BS116	3
CF125	Digital Electronics	BS114, UR112	3
CF127	Object Oriented Programming	CF 117	3
CF216	Data Structures	CF 127	3
CF225	Computer Architecture	CF125, CF117	3
CF229	Database Management Systems (1)	CF216	3
CF227	Operating Systems	CF125, CF127	3
CF223	Computer Networks	UR125, CF117	3
CF311	Web App Development	CF127	3
CF312	Data Visualization and Analytics	CF226, CF216	3
CF313	Software Engineering	CF216, CF229	3
CF321	Fundamental of Artificial Intelligence	CS312	3
Total			**39**

Faculty requirements of core computing courses are these courses offered to all students ofthe faculty programs by different departments which lead to further strong core science background for all computing & informatics students.

The weight of these courses is equivalent to **39** credit hours.

Article (34): Graduation Project

In the final two semesters, a senior student must be engaged in one of the research laboratories of the corresponding department to carry out his/her graduation project. The graduation project weights eight credit hours distributed on two semesters. One credit hour for the project is equivalent to four contact hours in labs/practical because the student needs to stay enough time in a research lab during the project period according to the facultybest practices. The student must earn at least 102 Credit Hours Level (including successfullypassing all of program's Level Three mandatory courses required by the project) to register for the graduation project. The first part of the graduation project must be registered first, while the second part of the graduation project can be registered the semester following that of the first part, without including summer term.

Article (35): University Regulations and Traditions and the Penalty for Violation

- Any violation of the laws, regulations and university traditions is considered a disciplinary offense, in particular:
- Acts that violate the system of the faculty or University facilities.
- Incitement to refrain from attending classes, exams, and other university work.
- Any act that is incompatible with good conduct and behavior inside or outside the University.
- Any violation of the exam system and any misconduct in the exam or attempt to do so.
- Every student who cheats in an exam or attempts to cheat in it is expelled by the dean or the chairman of the examination committee and is denied entry to the exam.
- Disciplinary sanctions are:
- Verbal or written warning.
- Reprimand letter.
- The denial of some student services.
- The deprivation of attending the lessons of one of the courses for a period not exceeding a month.
- Dismissal from the faculty for a period not to exceed one month.
- Prohibited from entering the final exam in one or more courses.
- Dismissal from faculty for a period not exceeding one semester.
- Dismissal from the University.
- The Dean of the faculty has to sign the first six sanctions after they are approved by the president of the University.
- The disciplinary council signs off on all sanctions and issues a decision to refer the student to the disciplinary council assigned by the president of the University at the request of the dean.
- The student disciplinary council shall be constituted as follows:

  Dean of the faculty.

  Vice-Dean of the Faculty.

  One of the faculty members chosen annually by the faculty council.

Department of Computer Science:
Computer Science Program
Undergraduate Program Specification

Program Mission

Computer Science (CS) Program advances the art of CS inspired by the problems arising in the field of computing sciences. The program trains future computer scientists in the theoretical / practical applications of this specialization. Computer Science is the central to all aspects of information technology.

Program intends to produce graduates as competent computer science practitionerswho have a solid foundation of basic and fundamental knowledge and experience in applying the existing technology to contemporary problems. This program well addresses the analytic skills required by students to develop their abilities in research and to proceed for postgraduate study.

Program Objectives

Computer Science program is designed to provide the student with the foundations ofthe discipline as well as the opportunity for specialization.

The Program Educational Objectives are:

a. To educate and train students in the fundamentals of computer science and mathematics, in order to analyze and solve computing problems, as demonstrated by their professional accomplishments in industry, government and graduate programs and measured within three to five years after graduation.
b. To educate students with an understanding of real-world computing needs, as demonstrated by their ability to address technical issues involving computing problems included in industry, government and graduate programs and measured within three to five years after graduation.
c. To train students to work effectively, professionally, and ethically in computing-related professions, as demonstrated by their communications, teamwork and leadership skills in industry, government and graduate programs and measuredwithin three to five years after graduation.

After successfully completing Computer Science program, graduate: Generally, in the computing and Information should be able to:

a. Apply fundamental concepts; techniques and latest technologies of computing, algorithms; systems development and networking to different types of real-world problems.
b. Able to evaluate systems requirements and user needs to develop integrated solutions.
c. Participate in all work processes and products to assure quality, accuracy and completeness within ICT environment.
d. Demonstrate professional responsibilities as well as ethical, societal and cultural concern with respect to ICT practices and usage.
e. Able to think clearly and logically with deep analytical skills to solve real world problems.
f. Contribute to the development and empowerment of his/her community.
g. Demonstrate effective communication; leadership; business administration and entrepreneurial skills.
h. Work collaboratively and engage in-group decision making within multi-disciplinary teams.
i. Able to analyze quantitative and qualitative data including financial, industry and scientific data.
j. Able to engage in life learning continuous professional development including post-graduate and research studies.

And specially, in the computer science program should be able to:

a. Apply different computer science themes, such as abstraction, complexity, and evolutionary change.
b. Apply mathematical foundations, algorithmic principles to develop computer applications for different platforms including embedded systems, high- performance clusters, and clouds.
c. Utilize a set of general principles, such as sharing a common resource, security issue, and concurrency.
d. Use different types of programming languages and tools to develop real world applications.

Typical Study Plan of CS – Program

Computer Science

Level 1

Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	UR111	Human Rights	حقوق الإنسان	None	1
	UR112	Introduction to Computers	مقدمة في الحاسبات	None	3
	BS113	English Language (1)	اللغة الإنجليزية (١)	None	3
	BS114	Physics and Electronics	فيزياء وإلكترونيات	None	3
	UR113	Entrepreneurship and Innovation	ريادة الأعمال والابتكار	None	1
	BS116	Mathematics (1): Calculus for Computing Sciences	الرياضيات (١): حساب التفاضل والتكامل لعلوم الحوسبة	None	3
	CF117	Introduction to Programming	مقدمة في البرمجة	None	3
Spring	UR122	Marketing & Digital Strategy	التسويق والإستراتيجية الرقمية	None	2
	CF123	Discrete Mathematics & Graph Theory	الرياضات المنفصلة و نظرية الأشكال	BS116	3
	BS124	English Language (2)	اللغة الإنجليزية (٢)	BS113	3
	CF125	Digital Electronics	الإلكترونيات الرقمية	BS114, UR112	3
	BS126	Mathematics (2): Linear Algebra & Applications	الرياضيات (٢): الجبر الخطي والتطبيقات	BS116	3
	CF127	Object Oriented Programming	البرمجة الشيئية	CF117	3

Level 2

Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	BS212	Mathematics (3): Differential Eq. and Fourier Transforms	الرياضيات (٣)	BS126	3
	CF225	Computer Architecture	معمارية الحاسب	CF125, CF117	3
	BS213	Project Management	إدارة المشروعات	None	3
	BS214	Probability and Statistics	الاحتمالية والإحصاء	BS116	3
	UR211	Communication Skills	مهارات الاتصال	None	2
	CF216	Data Structures	هياكل البيانات	CF127	3
Spring	UR221	Creative & Scientific Thinking	التفكير الإبداعي والعلمي	None	2
	UR222	Natural Resources & Sustainability	الموارد الطبيعية والاستدامة	None	1
	CF223	Computer Networks	شبكات الحاسب	UR112, CF117	3
	UR223	Research Methods and Ethics	طرق وأخلاقيات البحث العلمي	None	2
	CS230	Computer Graphics	رسومات الحاسب	CF216, BS126	3
	CF227	Operating Systems	نظم التشغيل	CF125, CF127	3
	CF229	Database Management Systems	نظم إدارة قواعد البيانات	CF216	3

Level 3					
Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	CS311	Introduction to Theoretical Computer Science	مقدمة في نظرية علوم الحاسب	CF225, CF227	3
	CF311	Web Application Development	تطوير تطبيقات الويب	CF223, CF229	3
	CF312	Data Visualization and Analytics	استعراض البيانات وتحليلها	CF216, CF229	3
	CF313	Software Engineering	هندسة البرمجيات	CF127	3
	CF321	Fundamentals of Artificial Intelligence	أساسيات الذكاء الاصطناعي	BS214, CF216	3
		Elective Course	مقرر اختياري		3
Spring	CS321	Theory of Algorithms	نظريات الخوارزميات	CS311	3
	CS315	Theory of Compilers	نظرية المترجمات	CF216	3
	CS322	Distributed Computer Systems	أنظمة الحاسبات الموزعة	CS311, CF227	3
	AI325	Image Processing & Pattern Recognition	معالجة الصور والتعرف على الأنماط	CF321	3
	CS324	Mobile Applications Development	تطوير تطبيقات الموبيل	CF127	3
		Elective Course	مقرر اختياري		3

Level 4					
Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	DI411	Computer and Cyber Security	الحاسب والأمن السيبراني	CF223	3
	CS411	Parallel Computing	المعالجة المتوازية	CS322	3
	CS412	Advanced Software Engineering	هندسة برمجيات متقدمة	CF313	3
		Elective Course	مقرر اختياري		3
	GP411	Project (1)	مشروع (١)	Article (40)	4
Spring	CS427	Embedded Systems	الأنظمة المدمجة	CF223, CF313	3
	DI423	Cloud Computing	الحوسبة السحابية	CS322	3
	CS422	Human Computer Interaction	واجهة الإنسان والحاسب	CF127	3
		Elective Course	مقرر اختياري		3
	GP412	Project (2)	مشروع (٢)	GP411	4

Computer Science													
Level 1								Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	UR111	Human Rights	حقوق الإنسان	None	1	1	0	0	20	40	-	40
	UR112	Introduction to Computers	مقدمة في الحاسبات	None	3	2	1	1	20	20	20	40
	BS113	English Language (1)	اللغة الإنجليزية (١)	None	3	2	2	0	20	40	-	40
	BS114	Physics and Electronics	فيزياء وإلكترونيات	None	3	2	0	2	20	20	20	40
	UR113	Entrepreneurship and Innovation	ريادة الأعمال والابتكار	None	1	1	0	0	20	40	-	40
	BS116	Mathematics (1): Calculus for Computing Sciences	الرياضيات (١): حساب التفاضل والتكامل لعلوم الحوسبة	None	3	2	2	0	20	40	-	40
	CF117	Introduction to Programming	مقدمة في البرمجة	None	3	2	0	2	20	20	20	40
Spring	UR122	Marketing & Digital Strategy	التسويق والإستراتيجية الرقمية	None	2	2	0	0	20	40	-	40
	CF123	Discrete Mathematics & Graph Theory	الرياضات المنفصلة و نظرية الأشكال	BS116	3	2	2	0	20	40	-	40
	BS124	English Language (2)	اللغة الإنجليزية (٢)	BS113	3	2	2	0	20	40	-	40
	CF125	Digital Electronics	الإلكترونيات الرقمية	BS114, UR112	3	2	0	2	20	20	20	40
	BS126	Mathematics (2): Linear Algebra & Applications	الرياضيات (٢): الجبر الخطي والتطبيقات	BS116	3	2	2	0	20	40	-	40
	CF127	Object Oriented Programming	البرمجة الشيئية	CF117	3	2	0	2	20	20	20	40

Level 2						Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	BS212	Mathematics (3): Differential Eq. and Fourier Transforms	الرياضيات (٣)	BS126	3	2	2	0	20	40	-	40
	CF225	Computer Architecture	معمارية الحاسب	CF125, CF117	3	2	1	1	20	20	20	40
	BS213	Project Management	إدارة المشروعات	None	3	2	2	0	20	40	-	40
	BS214	Probability and Statistics	الاحتمالية والإحصاء	BS116	3	2	2	0	20	40	-	40
	UR211	Communication Skills	مهارات الاتصال	None	2	2	0	0	20	40	-	40
	CF216	Data Structures	هياكل البيانات	CF127	3	2	0	2	20	20	20	40
Spring	UR221	Creative & Scientific Thinking	التفكير الإبداعي والعلمي	None	2	2	0	0	20	40	-	40
	UR222	Natural Resources & Sustainability	الموارد الطبيعية والاستدامة	None	1	1	0	0	20	40	-	40
	CF223	Computer Networks	شبكات الحاسب	UR112, CF117	3	2	0	2	20	20	20	40
	UR223	Research Methods and Ethics	طرق وأخلاقيات البحث العلمي	None	2	2	0	0	20	40	-	40
	CS230	Computer Graphics	رسومات الحاسب	CF216, BS126	3	2	0	2	20	20	20	40
	CF227	Operating Systems	نظم التشغيل	CF125, CF127	3	2	0	2	20	20	20	40
	CF229	Database Management Systems	نظم إدارة قواعد البيانات	CF216	3	2	0	2	20	20	20	40

Level 3						Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	CS311	Introduction to Theoretical Computer Science	مقدمة في نظرية علوم الحاسب	CF225, CF227	3	2	1	1	20	40	-	40
	CF311	Web Application Development	تطوير تطبيقات الويب	CF223, CF229	3	2	0	2	20	20	20	40
	CF312	Data Visualization and Analytics	استعراض البيانات وتحليلها	CF216, CF229	3	2	0	2	20	20	20	40
	CF313	Software Engineering	هندسة البرمجيات	CF127	3	2	1	1	20	20	20	40
	CF321	Fundamentals of Artificial Intelligence	أساسيات الذكاء الاصطناعي	BS214, CF216	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							
Spring	CS321	Theory of Algorithms	نظريات الخوارزميات	CS311	3	2	1	1	20	20	20	40
	CS315	Theory of Compilers	نظرية المترجمات	CF216	3	2	1	1	20	20	20	40
	CS322	Distributed Computer Systems	أنظمة الحاسبات الموزعة	CS311, CF227	3	2	0	2	20	20	20	40
	AI325	Image Processing & Pattern Recognition	معالجة الصور والتعرف على الأنماط	CF321	3	2	0	2	20	20	20	40
	CS324	Mobile Applications Development	تطوير تطبيقات الموبيل	CF127	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							

Level 4						Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	DI411	Computer and Cyber Security	الحاسب والأمن السيبراني	CF223	3	2	0	2	20	20	20	40
	CS411	Parallel Computing	المعالجة المتوازية	CS322	3	2	0	2	20	20	20	40
	CS412	Advanced Software Engineering	هندسة برمجيات متقدمة	CF313	3	2	1	1	20	20	20	40
		Elective Course	مقرر اختياري		3							
	GP411	Project (1)	مشروع (١)	Article (40)	4	2	0	4		60		40
Spring	CS427	Embedded Systems	الأنظمة المدمجة	CF223, CF313	3	2	0	2	20	20	20	40
	DI423	Cloud Computing	الحوسبة السحابية	CS322	3	2	0	2	20	20	20	40
	CS422	Human Computer Interaction	واجهة الإنسان والحاسب	CF127	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							
	GP412	Project (2)	مشروع (٢)	GP411	4	2	0	4		60		40

CS Degree Requirements

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
University Requirements	Mandatory	UR111	Human Rights	1	None	None	1	0	0
		UR112	Introduction to Computers	3	None	None	2	1	1
		UR113	Entrepreneurship and Innovation	1	None	None	1	0	0
		UR122	Marketing & Digital Strategy	2	None	None	2	0	0
		UR211	Communication Skills	2	None	None	2	0	0
		UR221	Creative & Scientific Thinking	2	None	None	2	0	0
		UR222	Natural Resources & Sustainability	1	None	None	1	0	0
		UR223	Research Methods and Ethics	2	None	None	2	0	0
			Total number of credit hours	14					

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
Basic Sciences	Mandatory	BS113	English Language (1)	3	None	None	2	2	0
		BS114	Physics and Electronics	3	None	None	2	0	2
		BS116	Mathematics (1): Calculus for Computing Sciences	3	None	None	2	2	0
		BS124	English Language (2)	3	BS113	English Language (1)	2	2	0
		BS126	Mathematics (2): Linear Algebra & Applications	3	BS116	Mathematics (1): Calculus for Computing Sciences	2	2	0
		BS212	Mathematics (3): Differential Eq. and Fourier Transforms	3	BS126	Mathematics (2): Linear Algebra & Applications	2	2	0
		BS213	Project Management	3	None	None	2	2	0
		BS214	Probability and Statistics	3	BS116	Mathematics (1): Calculus for Computing Sciences	2	2	0
			Total number of credit hours	24					

		Code	Course Title	Cr. Hr.	Pre-req		Lecture	Tutorial	Lab
Faculty Requirements	Mandatory	CF117	Introduction to Programming	3	None	None	2	0	2
		CF123	Discrete Mathematics & Graph Theory	3	BS116	None	2	2	0
		CF125	Digital Electronics	3	BS114, UR112	Physics and Electronics, Introduction to Computers	2	0	2
		CF127	Object Oriented Programming	3	CF117	Introduction to Programming	2	0	2
		CF216	Data Structures	3	CF127	Object Oriented Programming	2	0	2
		CF223	Computer Networks	3	UR112, CF117	Introduction to Computers, Introduction to Programming	2	0	2
		CF225	Computer Architecture	3	CF125, CF117	Digital Electronics, Introduction to Programming	2	1	1
		CF227	Operating Systems	3	CF125, CF127	Digital Electronics, Object Oriented Programming	2	0	2
		CF229	Database Management Systems	3	CF216	Data Structures	2	0	2
		CF311	Web Application Development	3	CF223, CF229	Computer Networks, Database Management Systems	2	0	2
		CF312	Data Visualization and Analytics	3	CF216, CF229	Data Structures, Database Management Systems	2	0	2
		CF313	Software Engineering	3	CF127	Object Oriented Programming	2	1	1
		CF321	Fundamentals of Artificial Intelligence	3	BS214, CF216	Probability and Statistics, Data Structures	2	0	2
			Total number of credit hours	39					

		Computer Science Program							
		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
Program Requirements	Mandatory	AI325	Image Processing & Pattern Recognition	3	CF321	Fundamentals of Artificial Intelligence	2	0	2
		CS230	Computer Graphics	3	CF216, BS126	Data Structures, Mathematics (2): Linear Algebra & Applications	2	0	2
		CS311	Introduction to Theoretical Computer Science	3	CF225, CF227	Computer Architecture, Operating Systems	2	1	1
		CS315	Theory of Compilers	3	CF216	Data Structures	2	1	1
		CS321	Theory of Algorithms	3	CS311	Introduction to Theoretical Computer Science	2	1	1
		CS322	Distributed Computer Systems	3	CS311, CF227	Introduction to Theoretical Computer Science, Operating Systems	2	0	2
		CS324	Mobile Applications Development	3	CF127	Object Oriented Programming	2	0	2
		CS411	Parallel Computing	3	CS322	Distributed Computer Systems	2	0	2
		CS412	Advanced Software Engineering	3	CF313	Software Engineering	2	0	2
		CS422	Human Computer Interaction	3	CF127	Object Oriented Programming	2	0	2
		CS427	Embedded Systems	3	CF223, CF313	Computer Networks, Software Engineering	2	0	2
		DI411	Computer and Cyber Security	3	CF223	Computer Networks	2	0	2
		DI423	Cloud Computing	3	CS322	Distributed Computer Systems	2	0	2
			Total number of credit hours	39					
	Electives	CS325	Software Component Design	3	CF216	Data Structures	2	0	2
		CS326	Software Verification	3	CF216	Data Structures	2	0	2
		CS327	Software Quality Assurance	3	CF216	Data Structures	2	0	2
		CS329	Gamification	3	CF321	Fundamentals of Artificial Intelligence	2	0	2
		CS330	Selected Topics in Computer Graphics	3	CF321	Fundamentals of Artificial Intelligence	2	0	2
		CS421	Types and Semantics for Programming Languages	3	CS311	Introduction to Theoretical Computer Science	2	0	2
		CS423	Parallel Programming	3	CF127	Object Oriented Programming	2	0	2
		CS424	3D Modeling	3	CF127, CS230	Object Oriented Programming, Computer Graphics	2	0	2
		CS425	Computer Animation	3	CF127, CS230	Object Oriented Programming, Computer Graphics	2	0	2

CS426	Multimedia	3	CF127, CS230	Object Oriented Programming, Computer Graphics	2	0	2
	Total number of credit hours	30					

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Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
GP411	Project (1)	4	Article (34)	Article (34)	2	0	4
GP412	Project (2)	4	GP411	Project (1)	2	0	4
	Total number of credit hours	8					

Code	Course Title	Cr. Hr.	Teaching			Mark Distribution %	
			Lecture	Tut	lab	Semester work	Oral Exam
FT300	Field Training	0	0	0	0	60%	40%

Department of Information Systems:
Information Systems Program
Undergraduate Program Specification

Program Mission

The program AIs to deliver high quality, accessible undergraduate information systems education to a diverse student population. The department prepares students to analyze, design, develop and use business applications utilizing contemporary technology. Also, a balancebetween the fundamental information systems concepts and the application of these conceptsfrom a future-oriented perspective is provided.

Program Objectives

a. To introduce students to relevant information technology, teamwork, and decision making necessary for the integration of business processes using information technology.
b. To provide students with understanding of the strategic use of information systems.
c. To prepare students to use the systems development life cycle along with its tools, techniques, methodologies, and processes, to develop computer-based information systems that meet the needs of businesses and organizations.
d. To equip students with the ability to analyze the motivations, architectures, and implementation of IS security policy technology and practices.
e. To provide students with web development and object-oriented programming skills to use in the development of solutions in business.
f. To provide students with database skills used to perform data analysis, design, implementation, and administration.
g. To equip students with a thorough understanding of the existing use of data communications networks and future developments in this area.
h. To prepare students for the development of computer-based systems projects fromthe point of request through analysis, design, development, documentation, testing, and implementation.

After successfully completing Computer Science program, graduate should be:

a. Apply fundamental concepts; techniques and latest technologies of computing, algorithms; systems development and networking to different types of real-world problems.
b. Able to evaluate systems requirements and user needs to develop integrated solutions.
c. Participate in all work processes and products to assure quality, accuracy and completeness within ICT environment.
d. Demonstrate professional responsibilities as well as ethical, societal and cultural concern with respect to ICT practices and usage.
e. Able to think clearly and logically with deep analytical skills to solve real worldproblems.
f. Contribute to the development and empowerment of his/her community.
g. Demonstrate effective communication; leadership; business administration and entrepreneurial skills.
h. Work collaboratively and engage in-group decision making within multi-disciplinaryteams.
i. Able to analyze quantitative and qualitative data including financial, industry and scientific data.
j. Able to engage in life learning continuous professional development including post-graduate and research studies.

And specially, in the information systems program should be able to:

a. Think and be creative through the programs offered and free studies included and provided students with research and debriefing skills enabling them to offer appropriate solutions in the field their specialty.
b. Provide students with the basic skills to prepare presentations, negotiate, and communicate so they can integrate different labor markets.
c. Recognize problems that could be solved via Information Systems.
d. Manage the storage and retrieval of digital data.
e. Design, create, manipulate, and implement computer-based information systems.
f. Familiarity with the ethical and legal standards and responsibilities of a profession

Typical Study Plan of IS – program

Information Systems

Level 1

Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	UR111	Human Rights	حقوق الإنسان	None	1
	UR112	Introduction to Computers	مقدمة في الحاسبات	None	3
	BS113	English Language (1)	اللغة الإنجليزية (١)	None	3
	BS114	Physics and Electronics	فيزياء وإلكترونيات	None	3
	UR113	Entrepreneurship and Innovation	ريادة الأعمال والابتكار	None	1
	BS116	Mathematics (1): Calculus for Computing Sciences	الرياضيات (١): حساب التفاضل والتكامل لعلوم الحوسبة	None	3
	CF117	Introduction to Programming	مقدمة في البرمجة	None	3
Spring	UR122	Marketing & Digital Strategy	التسويق والإستراتيجية الرقمية	None	2
	CF123	Discrete Mathematics & Graph Theory	الرياضات المنفصلة و نظرية الأشكال	BS116	3
	BS124	English Language (2)	اللغة الإنجليزية (٢)	BS113	3
	CF125	Digital Electronics	الإلكترونيات الرقمية	BS114, UR112	3
	BS126	Mathematics (2): Linear Algebra & Applications	الرياضيات (٢): الجبر الخطي والتطبيقات	BS116	3
	CF127	Object Oriented Programming	البرمجة الشيئية	CF117	3

Level 2

Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	BS212	Mathematics (3): Differential Eq. and Fourier Transforms	الرياضيات (٣)	BS126	3
	CF225	Computer Architecture	معمارية الحاسب	CF125, CF117	3
	BS213	Project Management	إدارة المشروعات	None	3
	BS214	Probability and Statistics	الاحتمالية والإحصاء	BS116	3
	UR211	Communication Skills	مهارات الاتصال	None	2
	CF216	Data Structures	هياكل البيانات	CF127	3
Spring	UR221	Creative & Scientific Thinking	التفكير الإبداعي والعلمي	None	2
	UR222	Natural Resources & Sustainability	الموارد الطبيعية والاستدامة	None	1
	CF223	Computer Networks	شبكات الحاسب	UR112, CF117	3
	UR223	Research Methods and Ethics	طرق وأخلاقيات البحث العلمي	None	2
	CS230	Computer Graphics	رسومات الحاسب	CF216, BS126	3
	CF227	Operating Systems	نظم التشغيل	CF125, CF127	3
	CF229	Database Management Systems	نظم إدارة قواعد البيانات	CF216	3

Level 3					
Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	DI311	Internet and Communication Technologies	تقنيات الإنترنت والاتصالات	CF223, CF227	3
	CF311	Web Application Development	تطوير تطبيقات الويب	CF223, CF229	3
	CF312	Data Visualization and Analytics	استعراض البيانات وتحليلها	CF216, CF229	3
	CF313	Software Engineering	هندسة البرمجيات	CF127	3
	CF321	Fundamentals of Artificial Intelligence	أساسيات الذكاء الاصطناعي	BS214, CF216	3
		Elective Course	مقرر اختياري		3
Spring	IS311	System Analysis and Design	تحليل وتصميم النظم	CF216, CF229	3
	IS325	Object Oriented Database	قاعدة بيانات كائنية	CF216	3
	IS322	Data Warehousing	مستودعات البيانات	CF229	3
	IS332	Data Sciences	علوم البيانات	BS214	3
	IS327	Information Retrieval	استرجاع المعلومات	CF229	3
		Elective Course	مقرر اختياري		3

Level 4					
Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	DI411	Computer and Cyber Security	الحاسب والأمن السيبراني	CF223	3
	IS434	Advanced Database Management Systems	إدارة نظم قواعد البيانات متقدم	CF229	3
	IS413	Big Data Analytics	تحليلات البيانات الضخمة	IS322	3
		Elective Course	مقرر اختياري		3
	GP411	Project (1)	مشروع (١)	Article (40)	4
Spring	IS435	Multimedia Information System	نظام معلومات الوسائط المتعددة	CF216	3
	IS425	Decision Support Systems	أنظمة دعم القرار	BS214	3
	IS432	Data Mining	استخراج البيانات	BS214	3
		Elective Course	مقرر اختياري		3
	GP412	Project (2)	مشروع (٢)	GP411	4

Typical Study Plan of IS – program

Information Systems													
Level 1						Teaching			Mark Dist. %				
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	UR111	Human Rights	حقوق الإنسان	None	1	1	0	0	20	40	-	40
	UR112	Introduction to Computers	مقدمة في الحاسبات	None	3	2	1	1	20	20	20	40
	BS113	English Language (1)	اللغة الإنجليزية (١)	None	3	2	2	0	20	40	-	40
	BS114	Physics and Electronics	فيزياء وإلكترونيات	None	3	2	0	2	20	20	20	40
	UR113	Entrepreneurship and Innovation	ريادة الأعمال والابتكار	None	1	1	0	0	20	40	-	40
	BS116	Mathematics (1): Calculus for Computing Sciences	الرياضيات (١): حساب التفاضل والتكامل لعلوم الحوسبة	None	3	2	2	0	20	40	-	40
	CF117	Introduction to Programming	مقدمة في البرمجة	None	3	2	0	2	20	20	20	40
Spring	UR122	Marketing & Digital Strategy	التسويق والإستراتيجية الرقمية	None	2	2	0	0	20	40	-	40
	CF123	Discrete Mathematics & Graph Theory	الرياضات المنفصلة و نظرية الأشكال	BS116	3	2	2	0	20	40	-	40
	BS124	English Language (2)	اللغة الإنجليزية (٢)	BS113	3	2	2	0	20	40	-	40
	CF125	Digital Electronics	الإلكترونيات الرقمية	BS114, UR112	3	2	0	2	20	20	20	40
	BS126	Mathematics (2): Linear Algebra & Applications	الرياضيات (٢): الجبر الخطي والتطبيقات	BS116	3	2	2	0	20	40	-	40
	CF127	Object Oriented Programming	البرمجة الشيئية	CF117	3	2	0	2	20	20	20	40

Level 2						Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	BS212	Mathematics (3): Differential Eq. and Fourier Transforms	الرياضيات (٣)	BS126	3	2	2	0	20	40	-	40
	CF225	Computer Architecture	معمارية الحاسب	CF125, CF117	3	2	1	1	20	20	20	40
	BS213	Project Management	إدارة المشروعات	None	3	2	2	0	20	40	-	40
	BS214	Probability and Statistics	الاحتمالية والإحصاء	BS116	3	2	2	0	20	40	-	40
	UR211	Communication Skills	مهارات الاتصال	None	2	2	0	0	20	40	-	40
	CF216	Data Structures	هياكل البيانات	CF127	3	2	0	2	20	20	20	40
Spring	UR221	Creative & Scientific Thinking	التفكير الإبداعي والعلمي	None	2	2	0	0	20	40	-	40
	UR222	Natural Resources & Sustainability	الموارد الطبيعية والاستدامة	None	1	1	0	0	20	40	-	40
	CF223	Computer Networks	شبكات الحاسب	UR112, CF117	3	2	0	2	20	20	20	40
	UR223	Research Methods and Ethics	طرق وأخلاقيات البحث العلمي	None	2	2	0	0	20	40	-	40
	CS230	Computer Graphics	رسومات الحاسب	CF216, BS126	3	2	0	2	20	20	20	40
	CF227	Operating Systems	نظم التشغيل	CF125, CF127	3	2	0	2	20	20	20	40
	CF229	Database Management Systems	نظم إدارة قواعد البيانات	CF216	3	2	0	2	20	20	20	40

Level 3						Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	DI311	Internet and Communication Technologies	تقنيات الإنترنت والاتصالات	CF223, CF227	3	2	1	1	20	20	20	40
	CF311	Web Application Development	تطوير تطبيقات الويب	CF223, CF229	3	2	0	2	20	20	20	40
	CF312	Data Visualization and Analytics	استعراض البيانات وتحليلها	CF216, CF229	3	2	0	2	20	20	20	40
	CF313	Software Engineering	هندسة البرمجيات	CF127	3	2	1	1	20	20	20	40
	CF321	Fundamentals of Artificial Intelligence	أساسيات الذكاء الاصطناعي	BS214, CF216	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							
Spring	IS311	System Analysis and Design	تحليل وتصميم النظم	CF216, CF229	3	2	1	1	20	20	20	40
	IS325	Object Oriented Database	قاعدة بيانات كائنية	CF216	3	2	0	2	20	20	20	40
	IS322	Data Warehousing	مستودعات البيانات	CF229	3	2	0	2	20	20	20	40
	IS332	Data Sciences	علوم البيانات	BS214	3	2	0	2	20	20	20	40
	IS327	Information Retrieval	استرجاع المعلومات	CF229	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							

Level 4						Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	DI411	Computer and Cyber Security	الحاسب والأمن السيبراني	CF223	3	2	0	2	20	20	20	40
	IS434	Advanced Database Management Systems	إدارة نظم قواعد البيانات متقدم	CF229	3	2	0	2	20	20	20	40
	IS413	Big Data Analytics	تحليلات البيانات الضخمة	IS322	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							
	GP411	Project (1)	مشروع (١)	Article (40)	4	2	0	4		60		40
Spring	IS435	Multimedia Information System	نظام معلومات الوسائط المتعددة	None	3	2	0	2	20	20	20	40
	IS425	Decision Support Systems	أنظمة دعم القرار	BS214	3	2	0	2	20	20	20	40
	IS432	Data Mining	استخراج البيانات	BS214	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							
	GP412	Project (2)	مشروع (٢)	GP411	4	2	0	4		60		40

IS Degree Requirements

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
University Requirements	Mandatory	UR111	Human Rights	1	None	None	1	0	0
		UR112	Introduction to Computers	3	None	None	2	1	1
		UR113	Entrepreneurship and Innovation	1	None	None	1	0	0
		UR122	Marketing & Digital Strategy	2	None	None	2	0	0
		UR211	Communication Skills	2	None	None	2	0	0
		UR221	Creative & Scientific Thinking	2	None	None	2	0	0
		UR222	Natural Resources & Sustainability	1	None	None	1	0	0
		UR223	Research Methods and Ethics	2	None	None	2	0	0
			Total number of credit hours	14					

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
Basic Sciences	Mandatory	BS113	English Language (1)	3	None	None	2	2	0
		BS114	Physics and Electronics	3	None	None	2	0	2
		BS116	Mathematics (1): Calculus for Computing Sciences	3	None	None	2	2	0
		BS124	English Language (2)	3	BS113	English Language (1)	2	2	0
		BS126	Mathematics (2): Linear Algebra & Applications	3	BS116	Mathematics (1): Calculus for Computing Sciences	2	2	0
		BS212	Mathematics (3): Differential Eq. and Fourier Transforms	3	BS126	Mathematics (2): Linear Algebra & Applications	2	2	0
		BS213	Project Management	3	None	None	2	2	0
		BS214	Probability and Statistics	3	BS116	Mathematics (1): Calculus for Computing Sciences	2	2	0
			Total number of credit hours	24					

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
Faculty Requirements	Mandatory	CF117	Introduction to Programming	3	None	None	2	0	2
		CF123	Discrete Mathematics & Graph Theory	3	BS116	None	2	2	0
		CF125	Digital Electronics	3	BS114, UR112	Physics and Electronics, Introduction to Computers	2	0	2
		CF127	Object Oriented Programming	3	CF117	Introduction to Programming	2	0	2
		CF216	Data Structures	3	CF127	Object Oriented Programming	2	0	2
		CF223	Computer Networks	3	UR112, CF117	Introduction to Computers, Introduction to Programming	2	0	2
		CF225	Computer Architecture	3	CF125, CF117	Digital Electronics, Introduction to Programming	2	1	1
		CF227	Operating Systems	3	CF125, CF127	Digital Electronics, Object Oriented Programming	2	0	2
		CF229	Database Management Systems	3	CF216	Data Structures	2	0	2
		CF311	Web Application Development	3	CF223, CF229	Computer Networks, Database Management Systems	2	0	2
		CF312	Data Visualization and Analytics	3	CF216, CF229	Data Structures, Database Management Systems	2	0	2
		CF313	Software Engineering	3	CF127	Object Oriented Programming	2	1	1
		CF321	Fundamentals of Artificial Intelligence	3	BS214, CF216	Probability and Statistics, Data Structures	2	0	2
			Total number of credit hours	39					

Information Systems Program

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
Program Requirements	Mandatory	IS311	System Analysis and Design	3	CF216, CF229	Data Structures, Database Management Systems	2	1	1
		IS322	Data Warehousing	3	CF229	Database Management Systems	2	0	2
		IS325	Object Oriented Database	3	CF216	Data Structures	2	0	2
		IS327	Information Retrieval	3	CF229	Database Management Systems	2	0	2
		IS332	Data Sciences	3	BS214	Probability and Statistics	2	0	2
		IS413	Big Data Analytics	3	IS322	Data Warehousing	2	0	2
		IS425	Decision Support Systems	3	BS214	Probability and Statistics	2	0	2
		IS432	Data Mining	3	BS214	Probability and Statistics	2	0	2
		IS434	Advanced Database Management Systems	3	CF229	Database Management Systems	2	0	2
		IS435	Multimedia Information System	3	None		2	0	2
		CS230	Computer Graphics	3	CF216, BS126	Data Structures, Mathematics (2): Linear Algebra & Applications	2	0	2
		DI311	Internet and Communication Technologies	3	CF223, CF227	Computer Networks, Operating Systems	2	1	1
		DI411	Computer and Cyber Security	3	CF223	Computer Networks	2	0	2
			Total number of credit hours	39					
	Electives	IS321	E-Commerce	3	UR113, UR122	Entrepreneurship and Innovation , Marketing & Digital Strategy	2	0	2
		IS323	Storage Technology	3	CF216	Data Structures	2	0	2
		IS324	Database Security	3	CF216	Data Structures	2	0	2
		IS325	Object Oriented Database	3	CF216	Data Structures	2	0	2
		IS326	Operation Research	3	BS214	Probability and Statistics	2	0	2
		IS328	Geographic Information System	3	CF216	Data Structures	2	0	2
		IS329	Healthcare Information System	3	CF216	Data Structures	2	0	2
		IS435	Multimedia Information System	3	CF216	Data Structures	2	0	2
		IS331	Business Process Management	3	IS311	System Analysis and Design	2	0	2
		IS412	Business Intelligence	3	CF321	Fundamentals of Artificial Intelligence	2	0	2
		IS422	Service-Oriented Architecture	3	CS322	Distributed Computer Systems	2	0	2
		IS424	Strategic Planning of Information Systems	3	None	None	2	0	2

IS425	Decision Support Systems	3	BS214	Probability and Statistics	2	0	2
IS426	Applied Information Theory	3	BS214	Probability and Statistics	2	0	2
IS427	Information Security	3	DI411, CF223	Computer and Cyber Security, Computer Networks	2	0	2
IS430	Knowledge Management	3	IS311	System Analysis and Design	2	0	2
IS431	Intelligence Information Systems	3	IS412	Business Intelligence	2	0	2
IS433	Mining Social Media	3	IS412	Business Intelligence	2	0	2
	Total number of credit hours	54					

	Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
Pr	GP411	Project (1)	4	Article (34)	Article (34)	2	0	4
	GP412	Project (2)	4	GP411	Project (1)	2	0	4
		Total number of credit hours	8					

Code	Course Title	Cr. Hr.	Teaching			Mark Distribution %	
			Lecture	Tut	lab	Semester work	Oral Exam
FT300	Field Training	0	0	0	0	60%	40%

Program Mission

To graduate a group of qualified individuals with the scientific knowledge and practical skills necessary to enable them to fulfill their responsibilities in the job sites that relateto business and activities in the field of industrial Interaction, and even to create appropriate applications for the country to achieve a qualitative shift in the field of automated applications.

Program Objectives

Our mission can be accomplished through:

a. Providing high quality teaching and research in accordance with international standards.
b. Mutual Interaction between teachers and students in a manner that supports the responsible good person.
c. Developing the capabilities of students to be able to think scientifically and to establish the method of scientific methods in solving their problems.
d. Directing the educational process and students' research towards contemporary problems and how to transform into an automated society.

After successfully completing Computer Science program, graduate should be:
Generally, in the computing and Information should be able to:

a. Apply fundamental concepts; techniques and latest technologies of computing, algorithms; systems development and networking to different types of real-world problems.
b. Able to evaluate systems requirements and user needs to develop integrated solutions.
c. Participate in all work processes and products to assure quality, accuracy and completeness within ICT environment.
d. Demonstrate professional responsibilities as well as ethical, societal and cultural concern with respect to ICT practices and usage.

e. Able to think clearly and logically with deep analytical skills to solve real world problems.
f. Contribute to the development and empowerment of his/her community.
g. Demonstrate effective communication; leadership; business administration and entrepreneurial skills.
h. Work collaboratively and engage in-group decision making within multi-disciplinary teams.
i. Able to analyze quantitative and qualitative data including financial, industry and scientific data.
j. Able to engage in life learning continuous professional development including post-graduate and research studies.

And specially, in the artificial intelligence program should be able to:

a. Apply the principles and techniques of data and knowledge representation, search, reasoning and learning with different data types.
b. Analyze real world problems, utilizing different problem-solving methods, and apply computational mechanisms.
c. Apply basics of mathematical science, computation theory, and modeling in solving real world problems.
d. Utilize fundamental and modern Artificial Intelligence techniques to develop computer applications in different domains.

Typical Study Plan of AI program

Artificial Intelligence

Level 1

Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	UR111	Human Rights	حقوق الإنسان	None	1
	UR112	Introduction to Computers	مقدمة في الحاسبات	None	3
	BS113	English Language (1)	اللغة الإنجليزية (١)	None	3
	BS114	Physics and Electronics	فيزياء وإلكترونيات	None	3
	UR113	Entrepreneurship and Innovation	ريادة الأعمال والابتكار	None	1
	BS116	Mathematics (1): Calculus for Computing Sciences	الرياضيات (١): حساب التفاضل والتكامل لعلوم الحوسبة	None	3
	CF117	Introduction to Programming	مقدمة في البرمجة	None	3
Spring	UR122	Marketing & Digital Strategy	التسويق والإستراتيجية الرقمية	None	2
	CF123	Discrete Mathematics & Graph Theory	الرياضات المنفصلة و نظرية الأشكال	BS116	3
	BS124	English Language (2)	اللغة الإنجليزية (٢)	BS113	3
	CF125	Digital Electronics	الإلكترونيات الرقمية	BS114, UR112	3
	BS126	Mathematics (2): Linear Algebra & Applications	الرياضيات (٢): الجبر الخطي والتطبيقات	BS116	3
	CF127	Object Oriented Programming	البرمجة الشيئية	CF117	3

Level 2

Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	BS212	Mathematics (3): Differential Eq. and Fourier Transforms	الرياضيات (٣)	BS126	3
	CF225	Computer Architecture	معمارية الحاسب	CF125, CF117	3
	BS213	Project Management	إدارة المشروعات	None	3
	BS214	Probability and Statistics	الاحتمالية والإحصاء	BS116	3
	UR211	Communication Skills	مهارات الاتصال	None	2
	CF216	Data Structures	هياكل البيانات	CF127	3
Spring	UR221	Creative & Scientific Thinking	التفكير الإبداعي والعلمي	None	2
	UR222	Natural Resources & Sustainability	الموارد الطبيعية والاستدامة	None	1
	CF223	Computer Networks	شبكات الحاسب	UR112, CF117	3
	UR223	Research Methods and Ethics	طرق وأخلاقيات البحث العلمي	None	2
	CS230	Computer Graphics	رسومات الحاسب	CF216, BS126	3
	CF227	Operating Systems	نظم التشغيل	CF125, CF127	3
	CF229	Database Management Systems	نظم إدارة قواعد البيانات	CF216	3

Level 3					
Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	AI311	Computational Cognitive Sciences	العلوم المعرفية الحاسوبية	CF216, BS126	3
	CF311	Web Application Development	تطوير تطبيقات الويب	CF223, CF229	3
	CF312	Data Visualization and Analytics	استعراض البيانات وتحليلها	CF216, CF229	3
	CF313	Software Engineering	هندسة البرمجيات	CF127	3
	CF321	Fundamentals of Artificial Intelligence	أساسيات الذكاء الاصطناعي	BS214, CF216	3
		Elective Course	مقرر اختياري		3
Spring	AI322	Natural Language Understanding and Generation	فهم اللغة الطبيعية وتوليدها	AI311	3
	AI326	Machine Learning	التعلم الآلي	CF321	3
	AI323	Intelligent Agents and Multi Agent Systems	الوكلاء الذكية وأنظمة الوكلاء المتعددة	CF216	3
	AI325	Image Processing & Pattern Recognition	معالجة الصور والتعرف على الأنماط	CF321	3
	AI331	Computer Vision	رؤية الكمبيوتر	BS212	3
		Elective Course	مقرر اختياري		3

Level 4					
Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	DI411	Computer and Cyber Security	الحاسب والأمن السيبراني	CF223	3
	AI411	Deep Learning	التعلم العميق	AI326	3
	AI412	Autonomous Systems	أنظمة مستقلة	AI311,	3
		Elective Course	مقرر اختياري		3
	GP411	Project (1)	مشروع (١)	Article (40)	4
Spring	AI421	Introduction to Mobile Robotics	مقدمة في الروبوتات المتنقلة	AI412	3
	AI429	Reinforcement Learning	التعلم التعزيزي	AI326	3
	CS427	Embedded Systems	الأنظمة المدمجة	BS114	3
		Elective Course	مقرر اختياري		3
	GP412	Project (2)	مشروع (٢)	GP411	4

Artificial Intelligence													
Level 1								Teaching			Mark Dist. %		
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	UR111	Human Rights	حقوق الإنسان	None	1	1	0	0	20	40	-	40
	UR112	Introduction to Computers	مقدمة في الحاسبات	None	3	2	1	1	20	20	20	40
	BS113	English Language (1)	اللغة الإنجليزية (١)	None	3	2	2	0	20	40	-	40
	BS114	Physics and Electronics	فيزياء وإلكترونيات	None	3	2	0	2	20	20	20	40
	UR113	Entrepreneurship and Innovation	ريادة الأعمال والابتكار	None	1	1	0	0	20	40	-	40
	BS116	Mathematics (1): Calculus for Computing Sciences	الرياضيات (١): حساب التفاضل والتكامل لعلوم الحوسبة	None	3	2	2	0	20	40	-	40
	CF117	Introduction to Programming	مقدمة في البرمجة	None	3	2	0	2	20	20	20	40
Spring	UR122	Marketing & Digital Strategy	التسويق والإستراتيجية الرقمية	None	2	2	0	0	20	40	-	40
	CF123	Discrete Mathematics & Graph Theory	الرياضات المنفصلة و نظرية الأشكال	BS116	3	2	2	0	20	40	-	40
	BS124	English Language (2)	اللغة الإنجليزية (٢)	BS113	3	2	2	0	20	40	-	40
	CF125	Digital Electronics	الإلكترونيات الرقمية	BS114, UR112	3	2	0	2	20	20	20	40
	BS126	Mathematics (2): Linear Algebra & Applications	الرياضيات (٢): الجبر الخطي والتطبيقات	BS116	3	2	2	0	20	40	-	40
	CF127	Object Oriented Programming	البرمجة الشيئية	CF117	3	2	0	2	20	20	20	40

Level 2						Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	BS212	Mathematics (3): Differential Eq. and Fourier Transforms	الرياضيات (٣)	BS126	3	2	2	0	20	40	-	40
	CF225	Computer Architecture	معمارية الحاسب	CF125, CF117	3	2	1	1	20	20	20	40
	BS213	Project Management	إدارة المشروعات	None	3	2	2	0	20	40	-	40
	BS214	Probability and Statistics	الاحتمالية والإحصاء	BS116	3	2	2	0	20	40	-	40
	UR211	Communication Skills	مهارات الاتصال	None	2	2	0	0	20	40	-	40
	CF216	Data Structures	هياكل البيانات	CF127	3	2	0	2	20	20	20	40
Spring	UR221	Creative & Scientific Thinking	التفكير الإبداعي والعلمي	None	2	2	0	0	20	40	-	40
	UR222	Natural Resources & Sustainability	الموارد الطبيعية والاستدامة	None	1	1	0	0	20	40	-	40
	CF223	Computer Networks	شبكات الحاسب	UR112, CF117	3	2	0	2	20	20	20	40
	UR223	Research Methods and Ethics	طرق وأخلاقيات البحث العلمي	None	2	2	0	0	20	40	-	40
	CS230	Computer Graphics	رسومات الحاسب	CF216, BS126	3	2	0	2	20	20	20	40
	CF227	Operating Systems	نظم التشغيل	CF125, CF127	3	2	0	2	20	20	20	40
	CF229	Database Management Systems	نظم إدارة قواعد البيانات	CF216	3	2	0	2	20	20	20	40

Level 3						Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	AI311	Computational Cognitive Sciences	العلوم المعرفية الحاسوبية	CF216, BS126	3	2	1	1	20	20	20	40
	CF311	Web Application Development	تطوير تطبيقات الويب	CF223, CF229	3	2	0	2	20	20	20	40
	CF312	Data Visualization and Analytics	استعراض البيانات وتحليلها	CF216, CF229	3	2	0	2	20	20	20	40
	CF313	Software Engineering	هندسة البرمجيات	CF127	3	2	1	1	20	20	20	40
	CF321	Fundamentals of Artificial Intelligence	أساسيات الذكاء الاصطناعي	BS214, CF216	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							
Spring	AI322	Natural Language Understanding and Generation	فهم اللغة الطبيعية وتوليدها	AI311	3	2	1	1	20	20	20	40
	AI326	Machine Learning	التعلم الآلي	CF321	3	2	0	2	20	20	20	40
	AI323	Intelligent Agents and Multi Agent Systems	الوكلاء الذكية وأنظمة الوكلاء المتعددة	CF216	3	2	1	1	20	20	20	40
	AI325	Image Processing & Pattern Recognition	معالجة الصور والتعرف على الأنماط	CF321	3	2	0	2	20	20	20	40
	AI331	Computer Vision	رؤية الكمبيوتر	BS212	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							

Level 4						Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	DI411	Computer and Cyber Security	الحاسب والأمن السيبراني	CF223	3	2	0	2	20	20	20	40
	AI411	Deep Learning	التعلم العميق	AI324	3	2	1	1	20	20	20	40
	AI412	Autonomous Systems	أنظمة مستقلة	AI311, AI312	3	2	1	1	20	20	20	40
		Elective Course	مقرر اختياري		3							
	GP411	Project (1)	مشروع (١)	Article (40)	4	2	0	4		60		40
Spring	AI421	Introduction to Mobile Robotics	مقدمة في الروبوتات المتنقلة	AI412	3	2	1	1	20	20	20	40
	AI429	Reinforcement Learning	التعلم التعزيزي	AI326	3	2	1	1	20	20	20	40
	CS427	Embedded Systems	الأنظمة المدمجة	CF223, CF313	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							
	GP412	Project (2)	مشروع (٢)	GP411	4	2	0	4		60		40

AI Degree Requirements

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
University Requirements	Mandatory	UR111	Human Rights	1	None	None	1	0	0
		UR112	Introduction to Computers	3	None	None	2	1	1
		UR113	Entrepreneurship and Innovation	1	None	None	1	0	0
		UR122	Marketing & Digital Strategy	2	None	None	2	0	0
		UR211	Communication Skills	2	None	None	2	0	0
		UR221	Creative & Scientific Thinking	2	None	None	2	0	0
		UR222	Natural Resources & Sustainability	1	None	None	1	0	0
		UR223	Research Methods and Ethics	2	None	None	2	0	0
			Total number of credit hours	14					

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
Basic Sciences	Mandatory	BS113	English Language (1)	3	None	None	2	2	0
		BS114	Physics and Electronics	3	None	None	2	0	2
		BS116	Mathematics (1): Calculus for Computing Sciences	3	None	None	2	2	0
		BS124	English Language (2)	3	BS113	English Language (1)	2	2	0
		BS126	Mathematics (2): Linear Algebra & Applications	3	BS116	Mathematics (1): Calculus for Computing Sciences	2	2	0
		BS212	Mathematics (3): Differential Eq. and Fourier Transforms	3	BS126	Mathematics (2): Linear Algebra & Applications	2	2	0
		BS213	Project Management	3	None	None	2	2	0
		BS214	Probability and Statistics	3	BS116	Mathematics (1): Calculus for Computing Sciences	2	2	0
			Total number of credit hours	24					

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
Faculty Requirements	Mandatory	CF117	Introduction to Programming	3	None	None	2	0	2
		CF123	Discrete Mathematics & Graph Theory	3	BS116	None	2	2	0
		CF125	Digital Electronics	3	BS114, UR112	Physics and Electronics, Introduction to Computers	2	0	2
		CF127	Object Oriented Programming	3	CF117	Introduction to Programming	2	0	2
		CF216	Data Structures	3	CF127	Object Oriented Programming	2	0	2
		CF223	Computer Networks	3	UR112, CF117	Introduction to Computers, Introduction to Programming	2	0	2
		CF225	Computer Architecture	3	CF125, CF117	Digital Electronics, Introduction to Programming	2	1	1
		CF227	Operating Systems	3	CF125, CF127	Digital Electronics, Object Oriented Programming	2	0	2
		CF229	Database Management Systems	3	CF216	Data Structures	2	0	2
		CF311	Web Application Development	3	CF223, CF229	Computer Networks, Database Management Systems	2	0	2
		CF312	Data Visualization and Analytics	3	CF216, CF229	Data Structures, Database Management Systems	2	0	2
		CF313	Software Engineering	3	CF127	Object Oriented Programming	2	1	1
		CF321	Fundamentals of Artificial Intelligence	3	BS214, CF216	Probability and Statistics, Data Structures	2	0	2
			Total number of credit hours	39					

Artificial Intelligence Program

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
Program Requirements	Mandatory	AI311	Computational Cognitive Sciences	3	CF216, BS126	Data Structures, Mathematics (2)	2	1	1
		AI322	Natural Language Understanding	3	AI311	Computational Cognitive Sciences	2	1	1
		AI323	Intelligent Agents and Multi-Agent Systems	3	CF216	Data Structures	2	1	1
		AI325	Image Processing & Pattern Recognition	3	CF321	Fundamentals of Artificial Intelligence	2	0	2
		AI326	Machine Learning	3	CF321	Fundamentals of Artificial Intelligence	2	0	2
		AI331	Computer Vision	3	BS212	Mathematics (3)	2	0	2
		AI411	Deep Learning	3	AI326	Neurosciences and Connectionist Networks Machine Learning	2	1	1
		AI412	Autonomous Systems	3	AI311	Computational Cognitive Sciences	2	1	1
		AI421	Introduction to Mobile Robotics	3	AI412	Autonomous Systems	2	1	1
		AI429	Reinforcement Learning	3	AI326	Machine Learning	2	1	1
		CS230	Computer Graphics	3	CF216, BS126	Data Structures, Mathematics (2)	2	0	2
		CS427	Embedded Systems	3	BS114	Physics and Electronics	2	0	2
		DI411	Computer and Cyber Security	3	CF223	Computer Networks	2	0	2
			Total number of credit hours	39					
	Electives	AI312	Computational Intelligence	3	CF216	Data Structures	2	1	1
		AI324	Neurosciences and Connectionist Networks	3	CF216	Data Structures	2	1	1
		AI327	Fundamental of Data Sciences	3	CF321	Fundamentals of Artificial Intelligence	2	0	2
		AI328	Probabilistic Reasoning	3	CF313, CF321	Software Engineering, Fundamentals of Artificial Intelligence	2	1	1
		AI329	Intelligence Multi Agent Systems	3	CF216	Data Structures	2	1	1
		AI330	Information Retrieval & Web Search	3	AI311	Computational Cognitive Sciences	2	1	1
		AI413	Statistical Artificial Intelligence	3	AI324	Neurosciences and Connectionist Networks	2	1	1
		AI422	AI and E-health	3	AI413	Statistical Artificial Intelligence	2	1	1

AI424	Natural Language Processing	3	AI322	Natural Language Understanding and Generation	2	1	1
AI425	Intelligence Autonomous Systems	3	CF321	Fundamentals of Artificial Intelligence	2	1	1
AI427	Forecasting and Predictive Analytics	3	AI326, AI327	Machine Learning, Fundamental of Data Sciences	2	1	1
AI428	Applied Data Sciences	3	AI326, AI327	Machine Learning, Fundamental of Data Sciences	2	1	1
AI430	Fuzzy Systems	3	CF321	Fundamentals of Artificial Intelligence	2	1	1
AI432	Programming for AI	3	CF127, CF216	Object Oriented Programming, Data Structures	2	0	2
AI433	Cognitive Robotics	5	IS432, AI425	Data Mining, Intelligence Autonomous Systems	2	0	2
AI434	Vision Intelligence of Robotics	3	IS432, AI425	Data Mining, Intelligence Autonomous Systems	2	0	2
AI435	Dynamics and Control of Robotics	3	IS432, AI425	Data Mining, Intelligence Autonomous Systems	2	0	2
AI437	Big Data Management and Modeling	3	IS432, AI425	Data Mining, Intelligence Autonomous Systems	2	0	2
AI438	Special Data Analysis	3	IS432, AI425	Data Mining, Intelligence Autonomous Systems	2	0	2
AI439	Social Network Analysis	3	IS432, AI425	Data Mining, Intelligence Autonomous Systems	2	0	2
AI440	Expert Systems	3	CF321	Fundamentals of Artificial Intelligence	2	1	1
AI441	Neural Networks	3	CF321	Fundamentals of Artificial Intelligence	2	1	1
AI442	Brain-Computer Interface	3	AI326, AI327	Machine Learning, Fundamental of Data Sciences	2	1	1
AI443	Generative AI	3	AI326, AI327	Machine Learning, Fundamental of Data Sciences	2	1	1
AI444	AI Ethics and Society	3	CF321	Fundamentals of Artificial Intelligence	2	2	0
	Total number of credit hours	74					

	Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
Pr	GP411	Project (1)	4	Article (40)	Article (40)	2	0	4
	GP412	Project (2)	4	GP411	Project (1)	2	0	4
		Total number of credit hours	8					

Code	Course Title	Cr. Hr.	Teaching			Mark Distribution %	
			Lecture	Tut	lab	Semester work	Oral Exam
FT300	Field Training	0	0	0	0	60%	40%

Department of Information Systems

Digital Informatics Program

Undergraduate Program Specification

Program Mission

Digital Informatics Program at the Faculty of Computers and InformationTechnology, Innovation University is committed to provide good education and research to create highly competitive cadres of specialists in the field of information technology, communications, and new technologies who are able to provide technical advice and solutions that contribute to the program in social and economic development.

Program Objectives

Digital Informatics Program at the Faculty of Computers and InformationTechnology-Innovation University is one of the best offered programs locally and regionally for academic/scientific research and applications in the field of Digital Informatics.

Its objectives are:

1- Preparing specialized and highly qualified human cadres in digital technology and its various fields.
2- Provide a high-quality program based on self-education and creative thinking with the application of performance appraisal systems and quality assurance.
3- Launching a specialized program in cooperation with specialists in the field, allowing the provision of modern technological specialties.
4- Offering a program based on the internationally accredited credit system which allows students to study according to their abilities, preferences, and interests.
5- Provide students with more space for practical training that meets the requirementsof work in all state institutions.
6- Highlight and enable scientifically distinguished students to achieve their scientific ambitions.

After successfully completing Digital Informatics program, graduate should be:

Generally, in the computing and Information, able to:

1. Apply fundamental concepts; techniques and latest technologies of computing, Internet, Communications, Mixed Reality, and Artificial Intelligent in sectors such as: Health, Energy, Construction, Transportation, Finance and Business.
2. Evaluate systems requirements and user needs to develop integrated solutions.
3. Participate in all work processes and products to assure quality, accuracy, and completeness.
4. Demonstrate professional responsibilities as well as ethical, societal, and cultural concern with respect to ICT practices and usage.
5. Think clearly and logically with deep analytical skills to solve real world problems.
6. Contribute to the development and empowerment of his/her community.
7. Demonstrate effective communication; leadership; business administration and entrepreneurial skills.
8. Work collaboratively and engage in-group decision making within multi-disciplinary teams.
9. Analyze quantitative and qualitative data including financial, industry and scientific data.
10. Engage in life learning continuous professional development including post-graduate and research studies.

And specially, in the digital informatics technology program, able to:

1. Identify methods and provide solutions which fuel the digital age that we live in.
2. Use digital methods and tools in the planning, analysis, design and management of complex systems and the Internet.
3. Develop digital twins and cyber-physical systems which power business and increase productivity and enhance management.
4. Think and be creative through the programs offered and free studies included and provided students with research and debriefing skills enabling them to offer appropriate solutions in the field their specialty.
5. Provide students with the basic skills to prepare presentations negotiate and communicate so they can integrate different labor markets.
6. Be Familiar with the ethical and legal standards and responsibilities of a profession.

Typical Study Plan of DI - program

Digital Informatics

Level 1

Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	UR111	Human Rights	حقوق الإنسان	None	1
	UR112	Introduction to Computers	مقدمة في الحاسبات	None	3
	BS113	English Language (1)	اللغة الإنجليزية (١)	None	3
	BS114	Physics and Electronics	فيزياء وإلكترونيات	None	3
	UR113	Entrepreneurship and Innovation	ريادة الأعمال والابتكار	None	1
	BS116	Mathematics (1): Calculus for Computing Sciences	الرياضيات (١): حساب التفاضل والتكامل لعلوم الحوسبة	None	3
	CF117	Introduction to Programming	مقدمة في البرمجة	None	3
Spring	UR122	Marketing & Digital Strategy	التسويق والإستراتيجية الرقمية	None	2
	CF123	Discrete Mathematics & Graph Theory	الرياضات المنفصلة و نظرية الأشكال	BS116	3
	BS124	English Language (2)	اللغة الإنجليزية (٢)	BS113	3
	CF125	Digital Electronics	الإلكترونيات الرقمية	BS114, UR112	3
	BS126	Mathematics (2): Linear Algebra & Applications	الرياضيات (٢): الجبر الخطي والتطبيقات	BS116	3
	CF127	Object Oriented Programming	البرمجة الشيئية	CF117	3

Level 2

Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	BS212	Mathematics (3): Differential Eq. and Fourier Transforms	الرياضيات (٣)	BS126	3
	CF225	Computer Architecture	معمارية الحاسب	CF125, CF117	3
	BS213	Project Management	إدارة المشروعات	None	3
	BS214	Probability and Statistics	الاحتمالية والإحصاء	BS116	3
	UR211	Communication Skills	مهارات الاتصال	None	2
	CF216	Data Structures	هياكل البيانات	CF127	3
Spring	UR221	Creative & Scientific Thinking	التفكير الإبداعي والعلمي	None	2
	UR222	Natural Resources & Sustainability	الموارد الطبيعية والاستدامة	None	1
	CF223	Computer Networks	شبكات الحاسب	UR112, CF117	3
	UR223	Research Methods and Ethics	طرق وأخلاقيات البحث العلمي	None	2
	CS230	Computer Graphics	رسومات الحاسب	CF216, BS126	3
	CF227	Operating Systems	نظم التشغيل	CF125, CF127	3
	CF229	Database Management Systems	نظم إدارة قواعد البيانات	CF216	3

Level 3

Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	DI311	Internet and Communication Technologies	تقنيات الإنترنت والاتصالات	CF223, CF227	3
	CF311	Web Application Development	تطوير تطبيقات الويب	CF117	3
	CF312	Data Visualization and Analytics	استعراض البيانات وتحليلها	CF216, CF229	3

	CF313	Software Engineering	هندسة البرمجيات	CF127	3
	CF321	Fundamentals of Artificial Intelligence	أساسيات الذكاء الاصطناعي	BS214, CF216	3
		Elective Course	مقرر اختياري		3
Spring	DI328	Network Management and Administration	إدارة الشبكة وإدارتها	CF223	3
	DI327	Multimedia Communication and Networking	اتصالات الوسائط المتعددة والشبكات	CF223	3
	DI329	Internet of Things	إنترنت الأشياء	CF223	3
	DI330	Mobile Networks	الشبكات المتحركة	CF223	3
	CS322	Distributed Computer Systems	أنظمة الحاسبات الموزعة	CS223, CF227	3
		Elective Course	مقرر اختياري		3

Level 4

Sem	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.
Fall	DI411	Computer and Cyber Security	الحاسب والأمن السيبراني	CF223	3
	DI412	Digital Twins	التوائم الرقمية	CF216, DI327	3
	DI413	Speech and Audio Signal Processing	معالجة الكلام والإشارات الصوتية	CF216	3
		Elective Course	مقرر اختياري	0	3
	GP411	Project (1)	مشروع (١)	Article (40)	4
Spring	DI422	Cyber Physical Systems	الأنظمة الفيزيائية السيبرانية	DI311, DI412	3
	CS422	Human Computer Interaction	واجهة الإنسان والحاسب	CF127	3
	DI423	Cloud Computing	الحوسبة السحابية	CS322	3
		Elective Course	مقرر اختياري	0	3
	GP412	Project (2)	مشروع (٢)	GP411	4

Digital Informatics

Level 1						Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	UR111	Human Rights	حقوق الإنسان	None	1	1	0	0	20	40	-	40
	UR112	Introduction to Computers	مقدمة في الحاسبات	None	3	2	1	1	20	20	20	40
	BS113	English Language (1)	اللغة الإنجليزية (١)	None	3	2	2	0	20	40	-	40
	BS114	Physics and Electronics	فيزياء وإلكترونيات	None	3	2	0	2	20	20	20	40
	UR113	Entrepreneurship and Innovation	ريادة الأعمال والابتكار	None	1	1	0	0	20	40	-	40
	BS116	Mathematics (1): Calculus for Computing Sciences	الرياضيات (١): حساب التفاضل والتكامل لعلوم الحوسبة	None	3	2	2	0	20	40	-	40
	CF117	Introduction to Programming	مقدمة في البرمجة	None	3	2	0	2	20	20	20	40
Spring	UR122	Marketing & Digital Strategy	التسويق والإستراتيجية الرقمية	None	2	2	0	0	20	40	-	40
	CF123	Discrete Mathematics & Graph Theory	الرياضات المنفصلة و نظرية الأشكال	BS116	3	2	2	0	20	40	-	40
	BS124	English Language (2)	اللغة الإنجليزية (٢)	BS113	3	2	2	0	20	40	-	40
	CF125	Digital Electronics	الإلكترونيات الرقمية	BS114, UR112	3	2	0	2	20	20	20	40
	BS126	Mathematics (2): Linear Algebra & Applications	الرياضيات (٢): الجبر الخطي والتطبيقات	BS116	3	2	2	0	20	40	-	40
	CF127	Object Oriented Programming	البرمجة الشيئية	CF117	3	2	0	2	20	20	20	40

Level 2						Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	BS212	Mathematics (3): Differential Eq. and Fourier Transforms	الرياضيات (٣)	BS126	3	2	2	0	20	40	-	40
	CF225	Computer Architecture	معمارية الحاسب	CF125, CF117	3	2	1	1	20	20	20	40
	BS213	Project Management	إدارة المشروعات	None	3	2	2	0	20	40	-	40
	BS214	Probability and Statistics	الاحتمالية والإحصاء	BS116	3	2	2	0	20	40	-	40
	UR211	Communication Skills	مهارات الاتصال	None	2	2	0	0	20	40	-	40
	CF216	Data Structures	هياكل البيانات	CF127	3	2	0	2	20	20	20	40
Spring	UR221	Creative & Scientific Thinking	التفكير الإبداعي والعلمي	None	2	2	0	0	20	40	-	40
	UR222	Natural Resources & Sustainability	الموارد الطبيعية والاستدامة	None	1	1	0	0	20	40	-	40
	CF223	Computer Networks	شبكات الحاسب	UR112, CF117	3	2	0	2	20	20	20	40
	UR223	Research Methods and Ethics	طرق وأخلاقيات البحث العلمي	None	2	2	0	0	20	40	-	40
	CS230	Computer Graphics	رسومات الحاسب	CF216, BS126	3	2	0	2	20	20	20	40
	CF227	Operating Systems	نظم التشغيل	CF125, CF127	3	2	0	2	20	20	20	40
	CF229	Database Management Systems	نظم إدارة قواعد البيانات	CF216	3	2	0	2	20	20	20	40

Level 3							Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	DI311	Internet and Communication Technologies	تقنيات الإنترنت والاتصالات	CF223, CF227	3	2	1	1	20	20	20	40
	CF311	Web Application Development	تطوير تطبيقات الويب	CF117	3	2	0	2	20	20	20	40
	CF312	Data Visualization and Analytics	استعراض البيانات وتحليلها	CF216, CF229	3	2	0	2	20	20	20	40
	CF313	Software Engineering	هندسة البرمجيات	CF127	3	2	1	1	20	20	20	40
	CF321	Fundamentals of Artificial Intelligence	أساسيات الذكاء الاصطناعي	BS214, CF216	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							
Spring	DI328	Network Management and Administration	إدارة الشبكة وإدارتها	CF223	3	2	0	2	20	20	20	40
	DI327	Multimedia Communication and Networking	اتصالات الوسائط المتعددة والشبكات	CF223	3	2	0	2	20	20	20	40
	DI329	Internet of Things	إنترنت الأشياء	CF223	3	2	0	2	20	20	20	40
	DI330	Mobile Networks	الشبكات المتحركة	CF223	3	2	0	2	20	20	20	40
	CS322	Distributed Computer Systems	أنظمة الحاسبات الموزعة	CS223, CF227	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							

Level 4						Teaching			Mark Dist. %			
Semester	Code	Title	Title in Arabic	Pre-Req	Cr. Hr.	Lecture	Tutorial	Lab	Midterm	Assignments	Lab	Final Exam
Fall	DI411	Computer and Cyber Security	الحاسب والأمن السيبراني	CF223	3	2	0	2	20	20	20	40
	DI412	Digital Twins	التوائم الرقمية	CF216, DI327	3	2	0	2	20	20	20	40
	DI413	Speech and Audio Signal Processing	معالجة الكلام والإشارات الصوتية	CF216	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							
	GP411	Project (1)	مشروع (١)	Article (40)	4	2	0	4		60		40
Spring	DI422	Cyber Physical Systems	الأنظمة الفيزيائية السيبرانية	DI311, DI412	3	2	0	2	20	20	20	40
	CS422	Human Computer Interaction	واجهة الإنسان والحاسب	CF127	3	2	0	2	20	20	20	40
	DI423	Cloud Computing	الحوسبة السحابية	CS322	3	2	0	2	20	20	20	40
		Elective Course	مقرر اختياري		3							
	GP412	Project (2)	مشروع (٢)	GP411	4	2	0	4		60		40

DI Degree Requirements

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
University Requirements	Mandatory	UR111	Human Rights	1	None	None	1	0	0
		UR112	Introduction to Computers	3	None	None	2	1	1
		UR113	Entrepreneurship and Innovation	1	None	None	1	0	0
		UR122	Marketing & Digital Strategy	2	None	None	2	0	0
		UR211	Communication Skills	2	None	None	2	0	0
		UR221	Creative & Scientific Thinking	2	None	None	2	0	0
		UR222	Natural Resources & Sustainability	1	None	None	1	0	0
		UR223	Research Methods and Ethics	2	None	None	2	0	0
			Total number of credit hours	14					

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
Basic Sciences	Mandatory	BS113	English Language (1)	3	None	None	2	2	0
		BS114	Physics and Electronics	3	None	None	2	0	2
		BS116	Mathematics (1): Calculus for Computing Sciences	3	None	None	2	2	0
		BS124	English Language (2)	3	BS113	English Language (1)	2	2	0
		BS126	Mathematics (2): Linear Algebra & Applications	3	BS116	Mathematics (1): Calculus for Computing Sciences	2	2	0
		BS212	Mathematics (3): Differential Eq. and Fourier Transforms	3	BS126	Mathematics (2): Linear Algebra & Applications	2	2	0
		BS213	Project Management	3	None	None	2	2	0
		BS214	Probability and Statistics	3	BS116	Mathematics (1): Calculus for Computing Sciences	2	2	0
			Total number of credit hours	24					

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
Faculty Requirements	Mandatory	CF117	Introduction to Programming	3	None	None	2	0	2
		CF123	Discrete Mathematics & Graph Theory	3	BS116	None	2	2	0
		CF125	Digital Electronics	3	BS114, UR112	Physics and Electronics, Introduction to Computers	2	0	2
		CF127	Object Oriented Programming	3	CF117	Introduction to Programming	2	0	2
		CF216	Data Structures	3	CF127	Object Oriented Programming	2	0	2
		CF223	Computer Networks	3	UR112, CF117	Introduction to Computers, Introduction to Programming	2	0	2
		CF225	Computer Architecture	3	CF125, CF117	Digital Electronics, Introduction to Programming	2	1	1
		CF227	Operating Systems	3	CF125, CF127	Digital Electronics, Object Oriented Programming	2	0	2
		CF229	Database Management Systems	3	CF216		2	0	2
		CF311	Web Application Development	3	CF117	Introduction to Programming	2	0	2
		CF312	Data Visualization and Analytics	3	CF216, CF229	Data Structures, Database Management Systems	2	0	2
		CF313	Software Engineering	3	CF127	Object Oriented Programming	2	1	1
		CF321	Fundamentals of Artificial Intelligence	3	BS214, CF216	Probability and Statistics, Data Structures	2	0	2
			Total number of credit hours	39					

		Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
Program Requirements	Mandatory	DI311	Internet and Communication Technologies	3	CF223, CF227	Computer Networks, Operating Systems	2	1	1
		DI327	Multimedia Communication and Networking	3	CF223	Computer Networks	2	0	2
		DI328	Network Management and Administration	3	CF223	Computer Networks	2	0	2
		DI329	Internet of Things	3	CF223	Computer Networks	2	0	2
		DI330	Mobile Networks	3	CF223	Computer Networks	2	0	2
		DI411	Computer and Cyber Security	3	CF223	Computer Networks	2	0	2
		DI412	Digital Twins	3	CF216, DI327	Data Structures, Multimedia Communication and Networking	2	0	2
		DI413	Speech and Audio Signal Processing	3	CF216	Data Structures	2	0	2
		DI422	Cyber Physical Systems	3	DI311, DI412	Internet and Communication Technologies, Digital Twins	2	0	2
		DI423	Cloud Computing	3	CS322	Distributed Computer Systems	2	0	2
		CS230	Computer Graphics	3	CF216, BS126	Data Structures, Mathematics (2): Linear Algebra & Applications	2	0	2
		CS322	Distributed Computer Systems	3	CF223, CF227	Computer Network, Operating Systems	2	0	2
		CS422	Human Computer Interaction	3	CF127	Object Oriented Programming	2	0	2
			Total number of credit hours	39					
	Electives	DI322	Cyber Ethics	3	CF216	Data Structures	2	2	0
		DI323	Storage Technology	3	CF216	Data Structures	2	0	2
		DI324	Network Automation	3	CF127, CF223	Object Oriented Programming, Computer Networks	2	0	2
		DI325	Network Routing & Switching	4	CF223	Computer Networks	2	0	2
		DI419	Wireless Systems Administration	3	DI325	Network Routing & Switching	2	0	2
		DI420	System Administration (1)	3	DI325	Network Routing & Switching	2	0	2
		DI424	Network Performance and Evaluation	3	CS322	Distributed Computer Systems	2	0	2
		DI425	Storage Area Networking	3	CS322	Distributed Computer Systems	2	0	2
		DI426	Grid Computing	3	CS322	Distributed Computer Systems	2	0	2
		DI427	Wireless Sensor Networks	3	CS322	Distributed Computer Systems	2	0	2
		DI428	System Administration (2)	3	DI420	System Administration (1)	2	0	2
		DI430	Digital Forensics	3	DI411	Computer and Cyber Security	2	0	2
		DI431	Ethical Hacking	3	DI411	Computer and Cyber Security	2	0	2

DI432	Network Crimes	3	DI411	Computer and Cyber Security	2	0	2
DI433	Internet Security	3	DI411	Computer and Cyber Security	2	0	2
	Total number of credit hours	46					

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Code	Course Title	Cr. Hr.	Pre-req	Pre-req Title	Lecture	Tutorial	Lab
GP411	Project (1)	4	Article (34)	Article (34)	2	0	4
GP412	Project (2)	4	GP411	Project (1)	2	0	4
	Total number of credit hours	8					

Code	Course Title	Cr. Hr.	Teaching			Mark Distribution %	
			Lecture	Tut	lab	Semester work	Oral Exam
FT300	Field Training	0	0	0	0	60%	40%

Course Catalog

Faculty of Computers and Information Technology

AI311	Computational Cognitive Sciences - العلوم المعرفية الحاسوبية
Pre-requisites	CF216, BS126: Data Structures, Mathematics (2): Linear Algebra & Applications
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	This course is an introduction to computational theories of human cognition. Drawing on formal models from classic and contemporary artificial intelligence, we will explore fundamental issues in human knowledge representation, inductive learning and reasoning. We will seek to answer the following questions: What are the forms that our knowledge of the world takes? What are the inductive principles that allow us to acquire new knowledge from the interaction of prior knowledge with observed data? What kinds of data must be available to human learners, and what kinds of innate knowledge (if any) must they have?

AI312	Computational Intelligence - الذكاء الحسابي
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	This course covers the fundamentals of Computational intelligence/Soft computing which includes Genetic Algorithm, Artificial Neural networks, Swarm optimization, Ant Colony, Fuzzy Systems, Bayes & Markov Networks, and Hybrid intelligent Systems like Adaptive Neuro-Fuzzy Inference System (ANFIS). It includes practical examples of the techniques implemented in Python

AI322	Natural Language Understanding and Generation - فهم اللغة الطبيعية وتوليدها
Pre-requisites	AI311: Computational Cognitive Sciences
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	This course provides a foundational understanding in natural language processing methods and strategies. Students will learn approaches to syntax and semantics in NLP, approaches to discourse, generation, dialogue, and summarization within NLP, understand current methods for statistical approaches to machine translation. They will also learn how to evaluate the strengths and weaknesses of various NLP technologies and frameworks as they gain practical experience in the NLP toolkits available. Students will also learn how to harness, employ, and analyze linguistic and textual data effectively.

AI323	Intelligent Agents and Multi Agent Systems - الوكلاء الذكية وأنظمة الوكلاء المتعددة
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	This course covers the concepts, applications, and theories of operations of Intelligent Agent Technology. An Intelligent Agent is a software program that uses communication protocols to exchange information for automatic problem solving. Students will perform an in-depth analysis of an Intelligent Agent for a specific application and construct a prototype of it. You will learn how Agents communicates together, across open and distributed environments, to solve a growing number of complex problems. The course also covers AI Planning.

AI324	Neurosciences and Connectionist Networks - علوم الأعصاب وشبكات الاتصال
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	This course will introduce (1) mathematical methods for neural-network analysis, providing a solid theoretical and analytical overview of the tools available to understand neural-network models; Multilayers perceptron, Hopfield network, Recurrent networks, and (2) key connectionist models with links to experimental observations, and which provide important targets for analytical results. It combines aspects of psychology, mathematics, and computer science. It uses computational models and artificial intelligence to further the understanding of the brain.

AI325	Image Processing & Pattern Recognition - معالجة الصور والتعرف على الأنماط
Pre-requisites	CF321: Fundamentals of Artificial Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	The objective of this course is to provide the fundamental concepts on digital image processing. Visual perception, image representation, image enhancement in spatial and frequency domain, image restoration, color image processing, image compression and finally, pattern recognition are the fields that will be covered during this course. At the end of this course, the student will be able to understand how digital images are treated and will acquire a basic knowledge of image processing.

AI326	Machine Learning - التعلم الآلي
Pre-requisites	CF321: Fundamentals of Artificial Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Concept of Supervised learning versus Unsupervised learning, Feature extraction, Data preprocessing, Dimensionality reduction, Classifiers and Prediction, Over fitting and under fitting, Decision tree and classification, Bayesian classifiers, K-Nearest neighbor, Linear and Logistic regression, Support Vector Machines (SVM), Hidden Markova model (HMM), Performance measures.

AI327	Fundamental of Data Sciences - أساسيات علوم البيانات
Pre-requisites	CF321: Fundamentals of Artificial Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Data science life cycle Introduction to Machine learning – Data Gathering – Going from data to insights and Value – Data Preparation – Model selection and building – Model evaluation- Introduction to data analytic – Introduction to data visualization. Formulate a Data –Driven Use Cases for data science projects – High level description and objectives of the use case – Challenges addressed by the use case – Pain points and impact of each challenge – Goals, success criteria, constraints and assumptions – Available data – Data sources and required resources – Modeling approach for each challenge – Overall model structure and workflow – Application of the use case into operational solution.

AI328	Probabilistic Reasoning - التفكير الاحتمالي
Pre-requisites	CF313, CF321: Software Engineering, Fundamentals of Artificial Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Bayesian data analysis; inference and networks; a disciplined approach to hypothesis formulation and test and to cultivate and appreciation of reasoning and decision making under uncertainty.

AI329	Intelligence Multi Agent Systems - الأنظمة الذكيه متعددة الوكلاء
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Intelligent Agents: Introduction – Search based Planning – Scaling Planning for Complex Tasks – Acting in Uncertain Environments – Algorithmic – game theoretic and logical foundations of multi-agent systems – Mathematical tools: Adjacency graph and matrix; Laplacian; Connectivity and Consensus; Passivity and Lyapunov stability; interconnection of mechanical systems – Multi-Agent architectures Intelligence communication – Cooperative distributed problem solving – Collaborative plans and social systems – Multi agent learning – Distributed Kalman filtering for sensor networks – Distributed rational decision making – Mutual localization with anonymous measurements; Target localization and encircling – Cooperative exploration of unknown environments – Applications in multi-robot control systems.

AI330	Information Retrieval & Web Search - استرجاع المعلومات والبحث على شبكة الإنترنت
Pre-requisites	AI311: Computational Cognitive Sciences
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Boolean and vector-Space Retrieval Models – Basic tokenizing, Text Indexing, Text similarity

AI331	Computer Vision - رؤية الكمبيوتر
Pre-requisites	BS212: Mathematics (3): Differential Eq. and Fourier Transforms
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	This course provides an introduction to computer vision including fundamentals of image formation, camera imaging geometry, feature detection and matching, stereo, motion estimation and tracking, image classification and scene understanding. We'll explore methods for depth recovery from stereo images, camera calibration, automated alignment, tracking, boundary detection, and recognition. We'll use both classical machine learning and deep learning to approach these problems. The focus of the course is to develop the intuitions and mathematics of the methods in lecture, and then to learn about the difference between theory and practice in the projects.

AI411	Deep Learning - التعلم العميق
Pre-requisites	AI324: Neurosciences and Connectionist Networks
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Performance evaluation of Information Retrieval Systems – Web Search, Automated Text, Categorization, Text Clustering – Text classification, and Recommender Systems

AI412	Autonomous Systems - أنظمة مستقلة
Pre-requisites	AI311, AI312: Computational Cognitive Sciences, Computational Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	The course covers all major aspects of control and systems engineering with an emphasis on system autonomy and intelligence. Learn how to build software for smart systems like driverless cars, smart robots, intelligent Chabot, and intelligent diagnostic systems. This course is designed to provide you with an in-depth understanding of the theory, methods, systems, software design and development used in Artificial intelligence and Autonomous Systems that can be used in different industries and organizations.

AI413	Statistical Artificial Intelligence - الذكاء الاصطناعي الإحصائي
Pre-requisites	AI324: Neurosciences and Connectionist Networks
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	This course introduces concepts and techniques in Artificial Intelligence that are grounded in Applied Statistics, including the fields of probability theory and big data analysis. Classical and Bayesian approaches are introduced to estimate parameters, quantify uncertainty, and test models in an Artificial Intelligence context. Techniques explored include Bayesian networks, regression, self-organizing maps, decision trees and ensemble methods. Explore how statistical techniques underpin machine learning models.

AI421	Introduction to Mobile Robotics - مقدمة في الروبوتات المتنقلة
Pre-requisites	AI412: Autonomous Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	This course provides practical programming and building of smart systems. Smart systems enclose the Robot Operating System (ROS), basic hardware and software modules to establish a concrete prototype, understanding of ROS, the design flow of autonomous and robotic systems and the ability to write modular and reusable libraries for robotic perception, control, motion, and planning. Students will be able to utilize different simulation environments with realistic models of robots, Basic concepts of robotic perception, mapping, and navigation.

AI422	AI and E-health - الذكاء الاصطناعي والصحة الإلكترونية
Pre-requisites	AI413: Statistical Artificial Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	This course applies methods and techniques learned from both modules; Fundamental AI and Machine Leaning in the field of healthcare. Students will learn how to analyze data on patient which includes medical and health records, medications prescribed, lab tests, and different procedures performed. It will also show how AI can help doctors diagnose patients more accurately, make predictions about patients' future health, and recommend better treatments.

AI424	Natural Language Processing - معالجة اللغة الطبيعية
Pre-requisites	AI322: Natural Language Understanding and Generation
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	The main objective of this course is to provide students with the field of computational linguistics and the theory and methods of natural language processing (NLP). We will learn how to create systems that can understand and produce human language, for applications such as information extraction, machine translation, automatic summarization, question-answering, and interactive dialogue systems. The course will cover linguistic (knowledge-based) and statistical approaches to language processing in the three major subfields on NLP: syntax (language structures), semantics (language meaning), and pragmatics/discourse (the interpretation of language in context). Analyzing and extracting information from large online corpora.

AI425	Intelligence Autonomous Systems - الانظمة الذكيه المستقلة
Pre-requisites	CF321: Fundamentals of Artificial Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	History of robotics – Actuators – Locomotion – Manipulation – Sensors and sensing for robotics – Sensor based and odometry Navigation – Workspace decomposition and search algorithms on graphs – Configuration Space and configuration-space obstacles – State estimation and dead reckoning – Bayesian filtering: Kalman and Particle Filters – Trajectory Planning – Sampling-based motion planning – PID control – Control using fuzzy logic – Hierarchical control – Simultaneous localization and mapping – Multi-robots and applications.

AI427	Forecasting and Predictive Analytics - التنبؤ والتحليلات التنبؤية
Pre-requisites	AI326, AI327: Machine Learning, Fundamental of Data Sciences
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Forecasting with predictive analytics offers the opportunity to leverage the huge amounts of data, now readily available, that exhibit the following characteristics: +Data that changes over time, as well as static data +Perhaps hundreds, thousands of independent variables +Data reflective of today's ever-changing economic environment With data mining and machine learning methods, your data can easily and quickly be converted into knowledge to yield more accurate and more actionable models. We will demonstrate these techniques on a data set from a Brazilian grocery store chain. Information including date, store, promotions, climate, and unit price will be used to predict the total sales of oranges for the company. The model will help you identify what affects sales, decide on best prices and promos, and most importantly, forecast sales in the future.

AI428	Applied Data Sciences - علوم البيانات التطبيقية
Pre-requisites	AI326, AI327: Machine Learning, Fundamental of Data Sciences
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	This course aims at introducing students to novel topics and applications in data science that need to be identified in a responsive manner as technology and needs evolve and develop including: block chain, health informatics, intelligence transportation, image recognition, search engines, natural language processing, etc.

AI429	Reinforcement Learning - التعلم التعزيزي
Pre-requisites	AI326: Machine Learning
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	This course introduces the fundamentals of reinforcement learning (RL), where agents learn to make decisions to maximize rewards. Students will explore key concepts such as Markov Decision Processes (MDPs), basic RL algorithms like Q-learning, and other advanced methods. The course includes lectures and hands-on assignments, with a final project applying RL techniques. Prerequisites are basic machine learning, a programming language like Python, and an understanding of linear algebra and probability.

AI430	Fuzzy Systems - أنظمة ضبابية
Pre-requisites	CF321: Fundamentals of Artificial Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Modern Theory of Fuzzy Logic, Fuzzy function approximation Fuzzification and de- fuzzification, Fuzzy additive systems, Fuzzy control and Chaos, Recursive Partitioning, Fuzzy filters for impulsive noise, Fuzzy decision, Fuzzy signal processing, Fuzzy image coding, Fuzzy communication, Fuzzy signal detection, Feed forward and feedback additive fuzzy systems, Fuzzy hardware.

AI432	Programming for AI - برمجة الذكاء الاصطناعي
Pre-requisites	CF127, CF216: Object Oriented Programming, Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Fundamental techniques and algorithms used for acquiring, processing, and extracting useful information from digital images. Emphasis will be placed on covering methods used for image sampling and quantization, image transforms, image enhancement and restoration, image encoding, image analysis and pattern recognition using Python or something like.

AI433	Cognitive Robotics - الروبوتات المعرفية
Pre-requisites	IS432, AI425: Data Mining, Intelligence Autonomous Systems
Credit Hours	5
Lecture, Tutorial, Lab	2, 0, 2
Description	Generic security models- HTTP Security- Proxy Servers and Firewalls- Cryptographic Techniques - oriented Security Protocols - SSL and TLS Protocols - Certificate Management and Public Key Infrastructures - Authentication and Authorization Infrastructures - Electronic Payment Systems- Client-side Security - Server-side Security- internet blocking and rating.

AI434	Vision Intelligence of Robotics - رؤية الذكاء من الروبوتات
Pre-requisites	IS432, AI425: Data Mining, Intelligence Autonomous Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	How do robots “see”, respond to and learn from their interaction with the world around them? This is the fascinating field of visual intelligence and machine learning. Visual intelligence allows a robot to “sense” and “recognize” the surrounding environment. It also enables a robot to “learn” from the memory of past experiences by extracting patterns in visual signals. You will understand how Machine Learning extracts statistically meaningful patterns in data that support classification, regression, and clustering. Then by studying Computer Vision and Machine Learning together you will be able to build recognition algorithms that can learn from data and adapt to new environments. By the end of this course, you will be able to program vision capabilities for a robot such as robot localization as well as object recognition using machine learning.

AI435	Dynamics and Control of Robotics - ديناميات والسيطرة على الروبوتات
Pre-requisites	IS432, AI425: Data Mining, Intelligence Autonomous Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	How can robots use their motors and sensors to move around in an unstructured environment? You will understand how to design robot bodies and behaviors that recruit limbs and more general appendages to apply physical forces that confer reliable mobility in a complex and dynamic world. We develop an approach to composing simple dynamical abstractions that partially automate the generation of complicated sensorimotor programs. Specific topics that will be covered include: mobility in animals and robots, kinematics and dynamics of legged machines, and design of dynamical behavior via energy landscapes.

AI437	Big Data Management and Modeling - إدارة البيانات الضخمة والنمذجة
Pre-requisites	IS432, AI425: Data Mining, Intelligence Autonomous Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to big data and its characteristics – Big data sources and applications – Introduction to virtualization and cloud computing – Map-Reduce paradigm – Hadoop Distributed File system (HDFS) – Data Locality for Hadoop in the Cloud Hadoop ecosystem (Hive – PIG – Spark – etc.) – Combined knowledge of Big Data technology and Data Science to understand how such combination is used to solve real-world applications. How Big Data and data science are used in modern business enterprises. Use cases of real problems will be addressed in this course.

AI438	Special Data Analysis - تحليل البيانات الخاصة
Pre-requisites	IS432, AI425: Data Mining, Intelligence Autonomous Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Analysis of different types of data including but not limited to: Spatial data (general concepts in spatial data analysis, analysis and modeling of point patterns, analyzing spatially continuous data, analysis of area data) – Temporal and hierarchical data (strings and sequences, w-grams and other approaches, time series, tree maps and hierarchies) – Sensor data – Structured, semi-structured and unstructured data analysis – Transactional data, etc.

AI439	Social Network Analysis - تحليل الشبكة الاجتماعية
Pre-requisites	IS432, AI425: Data Mining, Intelligence Autonomous Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Algorithms for analyzing online social networks, considering both their structure and coding - Fundamentals of social graph theory will be covered including distance, search, influence, community discovery, diffusion, and graph dynamics - Fundamentals of text analysis will also be covered with an emphasis on the type of text used in online social networks and common applications – Topics include sentiment classification, information extraction, clustering, and topic modeling - Influence and Centrality in Social Networks – Social ties and link prediction- Social Spam and Malicious Behavior prediction.

AI440	Expert Systems - نظم الخبراء
Pre-requisites	CF321: Fundamentals of Artificial Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Knowledge based expert systems – conventional programming versus knowledge engineering – knowledge acquisition and representation – inference methods – uncertainty models – languages and tools – fuzzy expert systems – knowledge engineering tools – applications.

AI441	Neural Networks - الشبكات العصبية
Pre-requisites	CF321: Fundamentals of Artificial Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Overview of neuro computing – neural network concepts – learning laws – associative networks – mapping networks – the back propagation neural network – self-organizing maps spatiotemporal, stochastic and hierarchical networks – packages for neural network applications.

AI442	Brain-Computer Interface - واجهة الدماغ والحاسوب
Pre-requisites	AI326, AI327: Machine Learning, Fundamental of Data Sciences
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Introduction to Brain-Computer Interface, Basic Neuroscience, Recording / Stimulation Techniques, Signal Processing and Machine Learning for BCI, Invasive BCIs, Neural Prosthetics, Decoding using Bayesian Filtering, Cognitive Control, Volitional control of neural activity and bidirectional neural Interfaces, Semi-Invasive BCIs with Electrocorticography (ECoG) – Nerve-Based Approaches – Evoked Potentials (SSVEP and P300) – Restoring Sensory and Motor Function – Security – Cognitive Monitoring – and Entertainment – BCI Applications in Robotic Avatars and Image Search.

AI443	Generative AI - الذكاء الاصطناعي التوليدي
Pre-requisites	AI326, AI327: Machine Learning, Fundamental of Data Sciences
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Generative models are widely used in many branches of artificial intelligence. This course covers mathematical and computational foundations of generative modeling, as well as applications in engineering, design, science, and the arts. Specific topics include variational autoencoders, generative adversarial networks, autoregressive models such as Transformers, normalizing flow models, information lattice learning, neural text decoding, prompt programming, and detection of generated content. Explainability and social responsibility in generative modeling will also be discussed, including topics of justice, human autonomy, and safety.

AI444	AI Ethics and Society - أخلاقيات الذكاء الاصطناعي والمجتمع
Pre-requisites	CF321: Fundamentals of Artificial Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 2, 0
Description	The course covers the interdisciplinary aspects of AI specifically how AI methods and systems interact and impact people, society, law and economy. It covers a selection of the following topics: explainable AI (XAI), fairness, algorithmic accountability and trust (FAT), artificial morality, responsible use of AI. It also offers practice in popular dissemination of AI.

BS100	Mathematics (0) -)0(الرياضيات
Pre-requisites	None
Credit Hours	0
Lecture, Tutorial, Lab	2, 0, 0
Description	Limit of one-variable functions – Continuity- Differentiability – Differentiation of Trigonometric functions, Exponential and Logarithmic functions, Inverse Trigonometric functions) – Applications of Differentiation (Extrema – Mean Value Theorem – Curve sketching – Related rates – L'Hopital's rule – Newton's method). Conic sections (Parabola hyperbola and ellipse) – Antiderivatives – Indefinite and Definite integrals – Integration techniques.

BS113	English Language (1) -)1(اللغة الإنجليزية
Pre-requisites	None
Credit Hours	3
Lecture, Tutorial, Lab	2, 2, 0
Description	This course gives practice in forming structured to consolidate and extend learners existing knowledge and give the students the knowledge about usage of correct English with an emphasis on reading and writing skills to be able to understand the information systems and computer science expressions and scientific terms.

BS114	Physics and Electronics - فيزياء وإلكترونيات
Pre-requisites	None
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	kinematics, dynamics, gravitation, work, energy, momentum, rotational kinematics, and dynamics. electric forces, electric fields, voltage, capacitance, current, resistance, magnetic forces, magnetic fields, induction, oscillations, and waves, principles of quantum mechanics, Electronic components and basic laws. Principles of circuits-analysis: dividers, equivalent sources, methods of solutions, circuits with nonlinear resistance, maximum power-transfer, sinusoidal excitation and impedance concept, magnitude and phase-shift of RLC circuits. Frequency response of linear circuits, passive filters types and characteristics. Diode-circuits: half and full-wave rectifiers, Zener regulators and limiters. Principles of transistor circuits and amplifiers.

BS116	Mathematics (1): Calculus for Computing Sciences - الرياضيات)1: حساب التفاضل والتكامل لعلوم الحوسبة)
Pre-requisites	None
Credit Hours	3
Lecture, Tutorial, Lab	2, 2, 0
Description	Function: continuity, limit, derivatives, mean value theorem and extreme values, graphing the function, optimization; Integration; technique of integrated, applications; Numerical differentiation and integration, applications using MATLAB, Python, or any other more preferred programming language. Series: tests for convergence, Taylor series. Multivariable function: partial derivatives.

BS124	English Language (2) -)2(اللغة الإنجليزية
Pre-requisites	BS113: English Language (1)
Credit Hours	3
Lecture, Tutorial, Lab	2, 2, 0
Description	This course aims to improve and develop the skills needed for the students to write effectively for academic purposes. It contains clear instruction and a range of activities to help them to understand what is required to plan, structuring, and write the assignments or reports. Understanding what writing assignment involves, identify strength and weakness, and understand the functions of essays and reports.

BS126	Mathematics (2): Linear Algebra & Applications - الجبر الخطي): 2(الرياضيات والتطبيقات
Pre-requisites	BS116: Mathematics (1): Calculus for Computing Sciences
Credit Hours	3
Lecture, Tutorial, Lab	2, 2, 0
Description	Vectors: scalar multiplication, dot product; Algebra of matrices and determinants: Invertible, power, polynomials in matrices, properties of determinants, applications to computer graphics; System of linear equations: Echelon form, rank, Gaussian elimination; Vector space: linear dependence and independence, basis and dimension, applications to Markov chains. Diagonalization: Cayley-Hamilton, eigenvalues and eigenvectors, discrete dynamical systems, applications to linear models.

BS211	Fundamentals of Management & Economics - أساسيات الإدارة والاقتصاد
Pre-requisites	None
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 0
Description	Managements concepts, characteristics of successful management, setting organization objectives and goals, organization, introduction to group psychology of organizations, organizational structure, Individuals and group dynamics, motivation, leadership, morale, power, labor-management behavior, people and production, control techniques and methods of human resources management, role of purchasing and materials management in business, Fundamental economic concepts and methods of economic analysis, economics and financing of IT business, Economic feasibility studies, Technical feasibility studies.

BS212	Mathematics (3): Differential Eq. and Fourier Transforms -)3(الرياضيات
Pre-requisites	BS126: Mathematics (2): Linear Algebra & Applications
Credit Hours	3
Lecture, Tutorial, Lab	2, 2, 0
Description	Ordinary differential equations: first order, higher order, homogeneous and nonhomogeneous. Fourier series, Fourier transform (FT), Discrete Fourier transform (DFT), Fast Fourier transform (FFT), applications in computer science using MATLAB, Interpolating polynomial and least squares

BS213	Project Management - إدارة المشروعات
Pre-requisites	None
Credit Hours	3
Lecture, Tutorial, Lab	2, 2, 0
Description	Project Management course provides learners with an integrative view of project management. It focuses on how projects contribute to the strategic goals of the organization. It enables learners to comprehend and apply a project plan's components, such as managing risk, scheduling resources, managing progress, and controlling projects. Topics covered include project selection and portfolio management, scope management, project team building, conflict and negotiation, cost estimation and budgeting, project scheduling, critical chain project scheduling, resource management, project evaluation, and control.

BS214	Probability and Statistics - الإحتمالية والإحصاء
Pre-requisites	BS116: Mathematics (1): Calculus for Computing Sciences
Credit Hours	3
Lecture, Tutorial, Lab	2, 2, 0
Description	Probability and Statistics for Computing sciences: random variables, basic probability theory, distributions and properties, sampling methods, EDA, estimation, hypothesis tests, regression, experimental design, transform methods, model construction, reliability, and joint distributions. Statistical inference; parametric models; point estimation; properties of estimators; maximum likelihood (ML) and properties of ML estimators; confidence intervals; hypothesis testing; goodness-of-fit tests; Bayesian inference; ANOVA; linear and logistic regression.

CF117	Introduction to Programming - مقدمة في البرمجة
Pre-requisites	UR112: Introduction to Computers
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Fundamental concepts of the procedural programming, Topics include algorithms and problem solving, data types, control structures, functions, arrays, files, and the mechanics of running, testing, and debugging. Basic syntax and semantics of a higher-level language, variables, types, expressions and assignment, simple I/O, conditional and iterative control structures, loop control structures – arrays, – Procedures and functions – functions and parameter passing, structured decomposition.

CF123	Discrete Mathematics & Graph Theory - الرياضات المنفصلة و نظرية الأشكال
Pre-requisites	BS116: None
Credit Hours	3
Lecture, Tutorial, Lab	2, 2, 0
Description	Propositional and predicate logic, valid arguments, methods of proof, Elementary set theory, Elementary graph theory, Relations & functions, Induction & recursive definitions, Counting methods (pigeonhole, inclusion/exclusion), introductory probability, Binary operations, groups, fields. Applications of finite fields, Elementary number theory: numeration, geometric topology, graph theory, design theory and other combinatorial ideas, Graphs, hyper graphs, transitive closure, trees, generating functions, recurrences, Planar Graphs, Colorability, Line Graphs.

CF125	Digital Electronics - الإلكترونيات الرقمية
Pre-requisites	BS114, UR112: Physics and Electronics, Introduction to Computers
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Binary number system and Boolean Algebra, Basic logic concepts, Minimization of logic functions, combinational and sequential digital circuits, minimization methods, state machine design and state assignment techniques, hardware description languages such as VHDL and Verilog, circuit implementation using integrated circuits and programmable logic devices.

CF127	Object Oriented Programming - البرمجة الشيئية
Pre-requisites	CF117: Introduction to Programming
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	The course covers principles of Structured programming, concepts, principles and methods of object-oriented programming using C# with .NET. Topics include classes, inheritance, polymorphism, interface, event handling and other object-oriented techniques. This course will also introduce C# programming with the .NET libraries, such as WinForms, ADO. NET, and ASP. NET.

CF216	Data Structures - هياكل البيانات
Pre-requisites	CF127: Object Oriented Programming
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Data structures, types, mapping of abstract information structures into representations on primary & secondary storage, Analysis of time & space complexity of algorithms, Sequences, Lists, Stacks, Queues, Sets, multi-sets, tables, Trees, Sorting. Hash tables. Priority queues. Graphs, String algorithms.

CF223	Computer Networks - شبكات الحاسب
Pre-requisites	UR112, CF117: Introduction to Computers, Introduction to Programming
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to basic networks concepts and implementation , data encapsulation - network topologies - network devices – transmission media - basic configuration of networks hardware including cabling, bridges, routers, and other communications. OSI & Internet reference models. Communication protocols for Local, Metropolitan & Wide Area Networks, BISDN networks, The Internet protocol suite, Mobile Networks, Quality of service in communication protocols, Network security, and Trends in communication networks.

CF225	Computer Architecture - معمارية الحاسب
Pre-requisites	CF125, CF117: Digital Electronics, Introduction to Programming
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Basic computer organization and design – Implementation of large memory systems – Cash memory – virtual memory – Bussing systems – control – DMA – I/O control methods – Interrupt – Synchronization – Alternative architectures: CISC and RISC, Pipeline and Vector processing, Multiprocessors.

CF227	Operating Systems - نظم التشغيل
Pre-requisites	CF125, CF127: Digital Electronics, Object Oriented Programming
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	The course covers operating systems (OS) overview, OS management hardware and software resources, OS components, OS functions, OS structure, processes management, deadlocks, memory management, virtual memory, secondary storage management, file system, production, securing the basic structure of computers, CPU structure, different micro- programming techniques, the ALU's structure and operation, the design of adders, subtractors, multipliers, memory types, memory design, and memory operations.

CF229	Database Management Systems - نظم إدارة قواعد البيانات
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	The course contents are database system concepts, design and architecture, concept of relational database, ER and enhanced ER models, Relational data model, Relational algebra, normalization techniques, SQL development, views, and schemas.

CF311	Web Application Development - تطوير تطبيقات الويب
Pre-requisites	CF223, CF229: Computer Networks, Database Management Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	The course focuses on designing and developing Web-based applications using a variety of programming languages and tools. The course includes Internet application development architecture, Class projects include developing business-to-consumer (B2C) and business-to-business (B2B) applications, create web pages using HTML specifically XHTML language, frames and form creation in HTML, design concepts in cascading style sheets, and web scripting with Java scripts, the primary client-side scripting language of the Internet

CF312	Data Visualization and Analytics - إستعراض البيانات وتحليلها
Pre-requisites	CF216, CF229: Data Structures, Database Management Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Data and information visualization is the graphical communication of data and information for the purposes of presentation, confirmation, exploration, and analysis. Images can be used to convey numbers, concepts, and relationships using techniques such as maps, icons, graphs, and other visual forms. In the past decade, visualization has evolved into a discipline, drawing from such fields as computer graphics, human-computer Interaction, perceptual psychology, and art. The emphasis of the course will be on exposing students to the current research issues and on identifying potential research topics in data visualization as it applies to large-scale big data systems.

CF313	Software Engineering - هندسة البرمجيات
Pre-requisites	CF127: Object Oriented Programming
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Software Engineering Process & Management: Generic view, Software Processes, Feasibility Studies, Project Management, Legal Aspects of Software Engineering, Requirements, Usability, System Architecture and Design, Object oriented Design, Reliability, Performance of Computer Systems, Business Aspects of Software Engineering, Risks in Software Development.

CF321	Fundamentals of Artificial Intelligence - أساسيات الذكاء الإصطناعي
Pre-requisites	BS214, CF216: Probability and Statistics, Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	This course has two parts; firstly, it introduces subtends to the world of AI. It focuses on the concept of symbolic representation and how to design AI system. It will focus on heuristics search, in particular A*, and defining the search space for some applications, games or otherwise. The second part introduces Description Logics, Ontology Web Language, sematic rules, and simple SPARQL queries. Students shall build ontology projects using Protégé.

CS230	Computer Graphics - رسومات الحاسب
Pre-requisites	CF216, BS126: Data Structures, Mathematics (2): Linear Algebra & Applications
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Fundamentals of random scan, raster-scan devices, LCD displays - point and line drawing techniques and algorithms, input/output devices and interactive techniques. Polygon filling methods: Seed fill, edge flag algorithm etc. - scan conversion techniques - anti aliasing techniques - clipping algorithms, Polygon clipping, Viewing transformation, Windowing transformation. Linear transformation: rotation, scaling, translation in 3D –homogeneous coordinates - normalized device coordinates - windowing and view porting, Cartesian Coordinates, Word view … etc. Curve generation - cubic splines, Beziers, blending of curves- other interpolation techniques, Displaying Curves and Surfaces, Shape description requirement, parametric function. Review of 3D vector algebra - parallel and perspective projections and transformation - hidden line/ surface elimination - shading and rendering - ray tracing techniques. Graphics software packages - segmentation and display files - graphics standards - graphics and computer networks - basic principles of X windows, X terminals, and Functions for segmenting display files.

CS311	Introduction to Theoretical Computer Science - مقدمة في نظرية علوم الحاسب
Pre-requisites	CF225, CF227: Computer Architecture, Operating Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	This course provides a first introduction to the theoretical concepts of Computer Science. The focus of the course is the study of abstract computing devices without targeting a specific programming language and/or computing platform. In particular, course contents are finite automata, which model computing machines with finite fixed memory, and the class of regular languages, which is used for pattern matching languages; pushdown automata and context-free grammars that facilitate declarative specifications of language syntax; the universal computational model of Turing machines, and the inherent limits of what can be solved on a computer (undesirability); and, time complexity theory, which helps to measure the used time to solve a problem.

CS315	Theory of Compilers - نظرية المترجمات
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Compiler modules; programming language specifications; lexical analysis, parsing - recursive descent & table driven; static semantics - symbol tables & type checking; error handling; introduction to code generation & optimization; compiler generators; interpreters, Introduction and overview, scanning-theory and practice – grammars and parsing – scanning- theory and practice: symbol tables – run-time storage organization – declaration processing - procedures and functions – code generation and optimization.

CS321	Theory of Algorithms - نظريات الخوارزميات
Pre-requisites	CS311: Introduction to Theoretical Computer Science
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	The course includes Analysis of algorithms, Solution of summation & recurrence equations, Algorithm paradigms such as divide-&-conquer, greedy algorithms, dynamic programming, backtracking, branch-&-bound, Advanced graph algorithms, Amortized analysis. Self-adjusting data structures, Complexity classes, NP-completeness, Approximation algorithms, Randomized algorithms.

CS322	Distributed Computer Systems - أنظمة الحاسبات الموزعة
Pre-requisites	CS311, CF227: Introduction to Theoretical Computer Science, Operating Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	The course includes Concepts of distributed computer systems architectures- Models of computation, Inter-process communication and middleware, Remote method invocation using Java RMI, Distributed object components using CORBA RMI, Fault tolerance, Time and Global states (Synchronization), Consistency and replication, Naming service, trading service, and Web Services.

CS324	Mobile Applications Development - تطوير تطبيقات الموبيل
Pre-requisites	CF127: Object Oriented Programming
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to mobile development, Mobile development platforms, web service, Mobile user interface, Mobile programming languages (pros and Cons) single platform vs. Cross- platform, Backend for mobile applications, Mobile API's, securing mobile applications: authentication, encryption, and data self-destruction, testing mobile applications, Mobile Marketing & Advertising.

CS325	Software Component Design - تصميم مكونات البرنامج
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Design and implementation of software subsystems, concept of design patterns is introduced, and common patterns are applied to the development of software components. Laboratory projects provide an opportunity for teams of students to implement components and to integrate them into complete systems. It introduces the use of Computer- Aided Software Engineering (CASE).

CS326	Software Verification - التحقق من البرامج
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Software testing, the Integration of testing into the software development process, Functional, combinational, structural, and model-based testing. Practical assignments will provide opportunities to apply software verification techniques and tools.

CS327	Software Quality Assurance - ضمان جودة البرمجيات
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Study of issues related to the uniqueness of software quality assurance (SQA). Topics include the environments for SQA methods, Software errors and failures, Software quality factors, SQA architecture, Contract review, Quality plans, Formal technical reviews, Software testing strategies and implementations, Automated testing tools, CASE tools and quality, Infrastructure component of SQA system.

CS329	Gamification - التلعيب
Pre-requisites	CF321: Fundamentals of Artificial Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Modern approaches to game programming such as game engine design, object modeling, game physics, Intelligence in games using AI, game networking, 2D & 3D graphics.

CS330	Selected Topics in Computer Graphics - مواضيع مختارة في رسومات الحاسوب
Pre-requisites	CF321: Fundamentals of Artificial Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Mondo Carlo integration, Markov Chain Mondo Carlo, Bounding Volume Hierarchies, Parallel Ray Tracing, Quasi- Mondo Carlo and Low-Discrepancy Patterns, Photon Mapping, Path Tracing, Metropolis Light Transport, Participating Media, Image Reconstruction (Denoising).

CS411	Parallel Computing - المعالجة المتوازية
Pre-requisites	CS322: Distributed Computer Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Principles of parallel algorithm design, Programming shared-address space systems, Parallel computer architectures, Programming scalable systems, Analytical modeling of program performance, Collective communication, Synchronization, Non-numerical algorithms, Numerical algorithms, Performance measurement and analysis of parallel programs

CS412	Advanced Software Engineering - هندسة برمجيات متقدمه
Pre-requisites	CF313: Software Engineering
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	The course aims to develop the broad understanding Software Modeling and Analysis. Internet fundamentals, concepts, and components in modern web, Markup language, Scripting language, Server-side scripting, Contents management system, Web hosting and security, Foundation, metrics, and measurement of software verification and validation, Review and static analysis, Testing, Problem Analysis, and reporting, Developing secure software.

CS421	Types and Semantics for Programming Languages - أنواع ودلالات لغات البرمجة
Pre-requisites	CS311: Introduction to Theoretical Computer Science
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Type systems and semantics are mathematical tools for precisely describing aspects of programming language. A type system imposes constraints on legal programs in order to guarantee their safe execution, whilst semantics specifies what a program will do when executed. This course gives an introduction to the main ideas and methods of type systems and semantics. This enables a deeper understanding of existing programming languages, as well as the ability to design and specify new language features. Inductive definitions and proof by induction, Untyped and simply-typed lambda calculus, Variable binding, Small step and big step semantics, Progress and preservation theorems, Products, sums, and list types, Reference types and exceptions, Subtyping, Subsumption and its understanding as inclusion or coercion, Principles of operational semantics, Key concepts of semantics: compositionality, adequacy, observational equivalence, full abstraction, and definability.

CS422	Human Computer Interaction - واجهة الإنسان والحاسب
Pre-requisites	CF127: Object Oriented Programming
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Human-computer Interaction spans interfaces from large wall-size computing down to handheld devices and invisible "ubiquitous computers". HCI teaches user-contents design thinking and methods for user studies. This course seals with Models of action, perception, cognition, and interaction in human-machine systems, Methods of Interaction analysis and interaction representation, Human-machine system evaluation, Practical implementation; Introduction to user and use-content design principles, Broader topics may include societal considerations, groupware, multimedia, media perspectives.

CS423	Parallel Programming - البرمجة المتوازية
Pre-requisites	CF127: Object Oriented Programming
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to Parallel Computing; Difference between Concurrent and Parallel, Distributed computing; Parallel Hardware and Parallel Software, SIMD and MIMD systems, coordinating the processes/threads, Parallel Program Design, Shared Memory Programming with OpenMP; Shared Memory Programming with P-threads; Performance evaluation metrics: throughput, Speedup and efficiency; Parallel Program Development.

CS424	3D Modeling - 3D النمذجة
Pre-requisites	CF127, CS230: Object Oriented Programming, Computer Graphics
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Basics of modeling and examines in depth the concepts, methods, and tools of polygonal modeling and NURBS (Non-uniform rational B-Spline) modeling in 3D; Acquainting with the basic terms of 3D modeling and components, three-dimensional modeling concepts. Experimenting with modeling using NURBS surfaces; Applying this geometry type, students create curves and surfaces to build up their models.

CS425	Computer Animation - الرسوم المتحركة الكمبيوتر
Pre-requisites	CF127, CS230: Object Oriented Programming, Computer Graphics
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Animation planning and production, traditional and emerging animation techniques, current animation topics, motion dynamics and particle systems, Developing a working knowledge of digital and traditional animation techniques and processes, a scholarly study of emerging advancements in computer animation technology will be included. The course also investigates the algorithms that make animations possible: key-framing, inverse kinematics, physical simulation, optimization, optimal control, motion capture, and data-driven methods.

CS426	Multimedia - الوسائط المتعددة
Pre-requisites	CF127, CS230: Object Oriented Programming, Computer Graphics
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Analog TV systems, raster representation of video, multiplexing of audio and video, frequency modulation, Color TV systems, YIQ color coordinate, multiplexing of YIQ and audio, de-multiplexing. Digital video basics: sampling and quantization, raw digital video format (CIF, BT601 and, HDTV), video format conversion: interlacing, NTSC/PAL conversion. Video coding: motion compensated temporal prediction, block matching algorithms (EBMA, HBMA), Video coding standards, digital TV systems, Multimedia, networks, multimedia applications RTP and RTCP, Audio and video streaming, internet phone and video conferencing, Multimedia programming.

CS427	Embedded Systems - الأنظمة المدمجة
Pre-requisites	CF223, CF313: Computer Networks, Software Engineering
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Embedded systems overview, Hardware / Software co-design, Examples of embedded systems, Components of Embedded systems – sensors, actuators, micro-controller processor Technology, IC technology, Issues in Design Technology. Scheduling Paradigms for real time systems - Static Priorities, Static Schedules, Dynamic Scheduling, Pre-emptive, Non-preemptive, Rate Monotonic, EDF, Real World Issues like - Task Assignment, CPU utilization, Blocking, Unpredictability, performance Measures. Hard & Soft Real Time Systems, Real Time Operating System concepts, RTOS services, capabilities, Resource Management. Programming Languages for Embedded Systems - Desired Language Characteristics. Tools for building Embedded systems, Embedded Software Development Methodology, Issues in real time databases, real time communications, Fault Tolerant Techniques – Fault Types, Detection, Recovery.

DI311	Internet and Communication Technologies - تقنيات الإنترنت والاتصالات
Pre-requisites	CF223, CF227: Computer Networks, Operating Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	The course covers the Computer networks terminology, architecture and topologies; basic terminology of Internet, Internet services, TCP/IP protocols; Web fundamentals and concepts; E-commerce; Communication through email and RSS feed; Fundamentals of cloud technologies, Classification of Google applications, Using Google applications; Web sites tools for planning, designing and development, Planning; and designing a web site – virtual concepts, interface, layout, navigation components; Web site classification, characteristics and purpose, Technologies for creating web sites, Tools for web site marketing, Publishing a web site online. Blog fundamentals and characteristics, Creating and maintaining a blog using popular platforms, Developing and managing a web site using an online software platform

DI322	Cyber Ethics - أخلاقيات الإنترنت
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 2, 0
Description	Ethics, morality, and privacy issues when working with technology, Topics include foundational and professional issues in cyber ethics; privacy, security, and crime in cyberspace; Intellection property and internet regulation; the digital divide and online communities; and emerging and converging technologies.

DI323	Storage Technology - تكنولوجيا التخزين
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Information and Data Storage, Development of Storage Technology, Intelligent Storage Components, RAID Technology, Common Storage Protocols, Storage System Architecture, Storage Resource Tuning and Data Protection Technology and Applications, Storage Business Continuity Solutions, Storage System Operation Management.

DI324	Network Automation - أتمتة الشبكة
Pre-requisites	CF127, CF223: Object Oriented Programming, Computer Networks
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to network automation - Scripting in PERL, Python, BASH, and Power shell to accomplish and automate common system administration tasks such as working with files, network and web communication, database interaction, and security.

DI325	Network Routing & Switching - توجيه الشبكة والتبديل
Pre-requisites	CF223: Computer Networks
Credit Hours	4
Lecture, Tutorial, Lab	2, 0, 2
Description	This course focuses on advanced IP addressing techniques, intermediate routing protocols, command-line interface configuration of switches, Ethernet switching, VLANs, STP, and VTP. Emphasis will be on application and demonstration of skills acquired in pre-requisite courses. Upon completion, students should be able to perform tasks related to VLSM, routing protocols, switching concepts and configuration, STP, VLANs, and VTP.

DI327	Multimedia Communication and Networking - اتصالات الوسائط المتعددة والشبكات
Pre-requisites	CF223: Computer Networks
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	The course is introducing students in multimedia communication systems and applications. The curriculum includes background knowledge in the areas of design and development of multimedia communication systems (digitizing, encoding, compression, transmission, analysis and mining of multimedia content) and the corresponding Internet technologies for streaming and Quality of Service for real-time multimedia communications. During the course case studies will be presented and there will be project assigned to students.

DI328	Network Management and Administration - إدارة الشبكة وإدارتها
Pre-requisites	CF223: Computer Networks
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	This course provides the theory and the practical skills necessary to understand and apply methods and tools for maintaining a computer network in operation. It offers in-depth study of emerging network management issues such as: network monitoring, network controlling, network security and programming network management applications.

DI329	Internet of Things - إنترنت الأشياء
Pre-requisites	CF223: Computer Networks
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to IoT-Physical and logical design of IoT-Engineering IoT networks; smart objects, connecting smart objects, IP as IoT network layer, application protocols for IoT, Data and analytics for IoT, securing IoT-Domains specific IoT-ethics and policy of IoT.

DI330	Mobile Networks - الشبكات المتحركة
Pre-requisites	CF223: Computer Networks
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Antennas: types, radiation patterns, antenna gain, propagation models, line of sight transmission, impairments, fading in wireless environment – satellites: orbits types, geostationary satellites, LEO satellites, MEO satellites, angles, footprint – capacity allocation - Cellular networks: definition, architecture, frequency reuse, mobile functions, handoff, power control – main architecture, last generation architecture – Ad-hoc networks: definition, problems, MAC protocol, routing, energy management.

DI411	Computer and Cyber Security - الحاسب والأمن السيبراني
Pre-requisites	CF223: Computer Networks
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Overview of security concepts and definitions, conventional classical and modern encryption algorithms (DES, AES), modern black ciphers and the data encryption standards, block cipher cryptanalysis and usage, modern stream ciphers, public key cryptography, key certificates and management, message authentication and hash functions, hash algorithms, digital signature and authentication protocols, electronic mail security, IP security, web security.

DI412	Digital Twins - التوائم الرقمية
Pre-requisites	CF216, DI327: Data Structures, Multimedia Communication and Networking
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	This course introduces the concept of Digital Twins, and it shows how it is applied in smart cities, manufacturing, and build environment. It will help in understanding how DTs can minimize the use of physical sensors on a real structure (asset), how to create a simulation model (analytical, FE or big data model) that replicates the physical asset. It will implement real example on how DTs can be utilized for predictive maintenance and decision support. A DT example can either be implemented as a (fast) local edge, a (slow) IoT cloud solution, or as a combination of both for maximum flexibility.

DI413	Speech and Audio Signal Processing - معالجة الكلام والإشارات الصوتية
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	This course teaches DSP basics, focusing on discrete signals and systems properties, the fundamental sequences, signal duration, periodic and aperiodic signals, symmetric sequence (even and odd parts), signal manipulations, signals decomposition and convolution. An introduction to automatic speech recognition (analysis and synthesis), the z transform, digital filters and discrete digital Fourier transform, and waves manipulations.

DI419	Wireless Systems Administration - إدارة الأنظمة اللاسلكية
Pre-requisites	DI325: Network Routing & Switching
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Wireless communications, standards, and regulations in an enterprise environment. Topics include various radio frequency and light communications; IEEE 802.11 Regulations and Standards; protocols and devices; network implementation; WAN security; and site surveying. WAN encapsulation.

DI420	System Administration (1) - إدارة النظام (1)
Pre-requisites	DI325: Network Routing & Switching
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	This course will introduce the tasks and techniques required to perform as a system administrator of modern operating systems. Topics to be covered include: booting, process control, the file system, managing users and resources, backups, configuration management, networking, the network file system, email servers, security, hardware devices, interoperability and daemons. Whenever possible, lectures will be augmented with hands-on exercises.

DI422	Cyber Physical Systems - الأنظمة الفيزيائية السيبرانية
Pre-requisites	DI311, DI412: Internet and Communication Technologies, Digital Twins
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Cyber Physical System bridges the cyber-world of computing and communications with the physical world as cyber-physical systems (CPS). This course covers important areas from the research literature on CPS. Three application domains are emphasized: medical devices for health care, smart transportation systems, and smart buildings. Several key cross-cutting principles, independent of the application domain, are also covered, including formal modeling, embedded systems, real-time systems, feedback control, and sensor networks.

DI423	Cloud Computing - الحوسبة السحابية
Pre-requisites	CS322: Distributed Computer Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to Cloud Computing, Cloud Computing Models, Compute Virtualization, Introduction to KVM, Network Basics for Cloud Computing, Physical Network for Virtualization, Virtual Network for Virtualization, Storage Virtualization Basics, Storage Architecture for Virtualization, Centralized Storage vs. Distributed Storage, Introduction to Virtualization Features, Cloud Computing Trends.

DI424	Network Performance and Evaluation - أداء الشبكة وتقييمها
Pre-requisites	CS322: Distributed Computer Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to NPE techniques, metrics and tools – sampling hypothesis testing – modeling-visual display of quantitative data – workload modeling – network management issues and tools – time Markov chain – single and multiple server query theory – operational analysis – simulation – verification and validation – input modeling – output analysis – comparing systems.

DI425	Storage Area Networking - شبكة منطقة التخزين
Pre-requisites	CS322: Distributed Computer Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Distributed network storage methods include ISCSI, DAS, NAS, and SAN technology. Other topics include configuration management, storage farms, backup, and recovery.

DI426	Grid Computing - الحوسبة الشبكية
Pre-requisites	CS322: Distributed Computer Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Middleware computing, grid monitoring: tasks, types, architecture, components, web and grid service, grid information system, grid resource management, grid scheduling, grid security system, grid enabled workflow system, semantic grid, knowledge grid.

DI427	Wireless Sensor Networks - شبكات الاستشعار اللاسلكية
Pre-requisites	CS322: Distributed Computer Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Sensor node hardware and operating system – configuration – applications – communication patterns - data gathering – sensor medium access control (MAC) protocol – localization – routing: indicator based, indicator free – power management – security issues.

DI428	System Administration (2) - إدارة النظام (2)
Pre-requisites	DI420: System Administration (1)
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Linux system installation and configuration in an enterprise environment, Topics include Linux system architecture; Linux installation and package management; GNU and UNIX Commands; Linux file systems; hierarchy standards; shells, scripting, and data management; user interfaces and desktops; administrative tasks; essential system services; and networks fundamentals and security.

DI430	Digital Forensics - الطب الشرعي الرقمي
Pre-requisites	DI411: Computer and Cyber Security
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction of the basic principles and technology of digital forensics, including acquisition, preservation, and recovery and investigation of the evidence stored in digital devices.

DI431	Ethical Hacking - القرصنة الأخلاقية
Pre-requisites	DI411: Computer and Cyber Security
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Foot printing and Reconnaissance, Scanning Networks, Enumeration, Vulnerability Analysis, System Hacking, Malware Threats, Sniffing, Social Engineering, Denial-of-Service, Session Hijacking, Evading IDS, Firewalls, and Honeypots, Hacking Web Servers, Hacking Web Applications, SQL Injection, Hacking Wireless Networks, Hacking Mobile Platforms, IoT Hacking, Cloud Computing, Cryptography.

DI432	Network Crimes - جرائم الشبكات
Pre-requisites	DI411: Computer and Cyber Security
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Computers and networks threats , network related crime cases: unauthorized access, theft of information, email bombing, data diddling, viruses, logic bombs, salami attacks, denial of services,…etc. , network crime preventive controls: transit security, traffic regulation – security actions: one level, two level, three level security actions , network crime detective controls: preparation, identification, containment, recovery and analysis , network crime corrective controls: before get started, regain control, analyze the intrusion , new technologies: finger scan technology, facial scan technology.

DI433	Internet Security - أمن الإنترنت
Pre-requisites	DI411: Computer and Cyber Security
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Generic security models- HTTP Security- Proxy Servers and Firewalls- Cryptographic Techniques - oriented Security Protocols - SSL and TLS Protocols - Certificate Management and Public Key Infrastructures - Authentication and Authorization Infrastructures - Electronic Payment Systems- Client-side Security - Server-side Security- internet blocking and rating.

GP411	Project (1) -)1(مشروع
Pre-requisites	Article (40): Article (40)
Credit Hours	4
Lecture, Tutorial, Lab	2, 0, 4
Description	The course requires that the student design, develop and implement a major project that solves a real problem in either business or engineering in the field of computer science. The general AI of the project is to allow the student to interact all the disciplines he has studied in a unified framework. On the behavioral side, students are allowed to work in a team so as to practice working in a collaborative environment. This emphasizes a proper documentation and presentation procedure. The project requires the presentation of oral and written reports.

GP412	Project (2) -)2(مشروع
Pre-requisites	GP411: Project (1)
Credit Hours	4
Lecture, Tutorial, Lab	2, 0, 4
Description	The course is a continuation of course FRM 416. The student completes the project begun in course FRM 416 and must pass a departmental assessment test. The student will have to take an external assessment examination approved by the department.

IS311	System Analysis and Design - تحليل وتصميم النظم
Pre-requisites	CF216, CF229: Data Structures, Database Management Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	This course presents a practical approach to do to perform the systems analysis and design to design, implement, and document the system development life cycle. It provides an avenue for solutions in the system through the various tasks involved in doing the analysis, Define and describe in detail the five phases of systems development life cycle (SDLC), Systems Planning, Systems Analysis; Systems Design; System Implementation; and Systems Operations and Support. The course Provides tools for communications, economic analysis, and Project planning skills across all phases of the SDLC.

IS321	E-Commerce - التجارة الإلكترونية
Pre-requisites	UR113, UR122: Entrepreneurship and Innovation , Marketing & Digital Strategy
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	This course emphasizes the major concepts and activities of e-commerce. It presents an e-commerce framework. Various topics are covered including electronic payment systems, order fulfillment, electronic customer relationship management, e-auctions, business to business e-commerce, selling and buying in private e-markets, e-supply chains, collaborative commerce, intra business ecommerce, corporate portals, mobile commerce and pervasive computing.

IS322	Data Warehousing - مستودعات البيانات
Pre-requisites	CF229: Database Management Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to data warehouses – Data warehouses and decision support systems – data warehouses characteristics and architecture – data warehouse tools – Design issues – Dimensional models and data cubes – Data warehouse models: star schema and snowflake schema – Data warehouses in real world: Sales – Inventory – Education – Health – Transportation – and others – Querying data warehouses using MDX.

IS323	Storage Technology - تكنولوجيا التخزين
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Information and Data Storage, Development of Storage Technology, intelligent Storage Components, RAID Technology, Common Storage Protocols, Storage System Architecture, Storage Resource Tuning and Data Protection Technology and Applications, Storage Business Continuity Solutions, Storage System Operation Management.

IS324	Database Security - أمن قاعدة البيانات
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to database security, Access control management features, Cryptographic data protection Database security methods and techniques, Access control to database objects: tables, attributes, records, Cryptographic methods of protection, Data

IS325	Object Oriented Database - قاعدة بيانات كائنية
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Data models, Semantic data models, Problems in record-oriented models, Object data model, Classes and inheritance, Methods and messages, Multiple inheritance, Object queries, Object query language OQL, indexing in object databases, Processing object queries, Object transactions, Concurrency control in object databases, Security in object databases, Using the object model in advanced applications.

IS326	Operation Research - بحوث العمليات
Pre-requisites	BS214: Probability and Statistics
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Project management: Network representation. Critical Path Method (CPM), Construction of time scheduling, Program Evaluation and Review Technique (PERT), Decision Analysis: Decision making under certainty, decision making under risk , and decision making under uncertainty, Inventory Models: Economic Order Quantity (EOQ) model, non-instantaneous replenishment model, and shortage model, Stochastic Processed and Markov Chain, Queuing models: exponential and Poisson distributions, basic structure of queuing models (single and multiple server).

IS327	Information Retrieval - استرجاع المعلومات
Pre-requisites	CF229: Database Management Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Intelligent Information Retrieval and Web Search Systems- Information Retrieval System Architecture- - Information Retrieval System Components - Boolean Retrieval- Inverted Index- Boolean and Vector Space Retrieval Models-Query Operations: Relevance Feedback & Query Expansion- Performance Evaluation of Information Retrieval Systems- Basic Tokenizing, Indexing, and Implementation of Vector-Space Retrieval- Term Vocabulary and Postings Lists- Dictionaries and Tolerant Retrieval.

IS328	Geographic Information System - نظام المعلومات الجغرافية
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Defining GIS, Components of a GIS, How GIS work, Map projections and spatial reference systems, GIS data structure, Modeling real word (Raster, Vector), Types of maps as visualization tool, Mobile GIS, GIS remote sensing, Geo database, GIS challenges, Global Positioning System (GPS), GPS challenges, GIS Applications.

IS329	Healthcare Information System - نظام معلومات الرعاية الصحية
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Understanding HIT, Types of Healthcare Facilities, Departments in the Hospital, Data Flow in HIS; Radiology DF, Laboratory DF, Surgery DF, Pharmacy DF, Organizational Behavior; Handling protected health information (PHI), Identifying EHR/EMR Access Roles and Responsibilities, Medical Business Operations, Secure Medical Operations, Regulatory Requirements and Ethics.

IS331	Business Process Management - إدارة العمليات التجارية
Pre-requisites	IS311: System Analysis and Design
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to Business Process Management (BPM) – Business Process Lifecycle – Process Modeling Techniques – Business Process Model and Notation – Petri nets – Correctness of Business Process Models – Simulation of Business Process Models – Process Instantiation Semantics – BPEL – Mapping from BPMN to BPEL.

IS332	Data Sciences - علوم البيانات
Pre-requisites	BS214: Probability and Statistics
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Knowledge discovery in databases, Data mining process, Data cleaning and preparation, Mining association rules, Classification, Prediction, Clustering, Web mining, Applications of data mining, Mining advanced databases. Business analytics is about using data, analytics, and algorithms to make prescriptive predictions about future events and decisions. This course will take a holistic approach to helping participants understand the key factors involved, from data collection to analysis to prediction and insight. Projects will give students hands-on experience developing and running a data science pipeline to ensure that the correct business predictions are being made. Emphasis will be on merging technical skills with critical thinking to ensure that robust data science pipelines are being created for business benefit.

IS412	Business Intelligence - ذكاء الأعمال
Pre-requisites	CF321: Fundamentals of Artificial Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Business Intelligence (BI) refers to technologies, applications, and practices for the collection, integration, analysis, and presentation of business information. The purpose of business intelligence is to support better business decision making. This course provides an overview of the technology of BI and the application of BI to an organization's strategies and goals.

IS413	Big Data Analytics - تحليلات البيانات الضخمة
Pre-requisites	IS322: Data Warehousing
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Big Data Industry and Technological Trends, Hadoop Distributed File System, MapReduce , In- memory Distributed Computing Engine (Spark2x),Distributed NoSQL Database (HBase), Distributed Data Warehouse (Hive), Streaming , Stream Processing and Batch Processing Platform (Flink), Data Transformation (Loader), Massive Logs Aggregation (Flume), Distributed Message Subscription System (Kafka), Cluster Distributed Coordination Service (Zookeeper).

IS422	Service-Oriented Architecture - البنية الخدميه
Pre-requisites	CS322: Distributed Computer Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Evolution of client server computing in the codex of the World Wide Web – Design and develop a functioning distributed application: various standards and specifications that support service orientation – XML – API's – DBMS APIs – XML databases and XQuery – Directory Services – Application Servers and Registries – Security and configuration issues for enterprise systems – Design issues for n-tier distributed systems.

IS424	Strategic Planning of Information Systems - التخطيط الاستراتيجي لنظم المعلومات
Pre-requisites	None
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to IS strategic Planning – Business strategy concepts – IS Governance- Developing an IS strategy, understanding the current situation, determining the future potentials, determining the business information strategy – Managing the information resource – Managing the supply of IT applications and Infrastructure.

IS425	Decision Support Systems - أنظمة دعم القرار
Pre-requisites	BS214: Probability and Statistics
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Distributed network storage methods include ISCSI, DAS, NAS, and SAN technology. Other topics include configuration management, storage farms, backup, and recovery.

IS426	Applied Information Theory - نظرية المعلومات التطبيقية
Pre-requisites	BS214: Probability and Statistics
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	The basic concepts of Information Theory: entropy, relative entropy, mutual information, channel capacity, and rate distortion (data compression), Applications of these concepts will be emphasized, The notions of sparse-graph codes, message passing, and belief propagation, Applications in inference algorithms and machine learning, topics of neural networks, statistical estimation, community detection and clustering.

IS427	Information Security - أمن المعلومات
Pre-requisites	DI411, CF223: Computer and Cyber Security, Computer Networks
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to Information security- threats to Information systems - planning for information security-Information risk management-information security governance- Incident management- controlling computers and files – Authenticating people- encrypting files and volumes – Information security maintenance.

IS430	Knowledge Management - إدارة المعرفة
Pre-requisites	IS311: System Analysis and Design
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to Knowledge Management, Knowledge Managements Activities, Nonaka's Model, Knowledge Taxonomy, The Learning Organization, Knowledge Repository, Knowledge Management Life cycle, Design of the KM Blueprint, Knowledge engineering, Knowledge acquisition, Knowledge Representation, Knowledge Validation, Knowledge Inference, Expert Systems.

IS431	Intelligence Information Systems - نظم المعلومات الذكية
Pre-requisites	IS412: Business Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Knowledge based systems, expert systems, concepts and definition of AI, machine learning techniques and methods, case-based reasoning, fuzzy expert systems, support vector machine, intelligence agents.

IS432	Data Mining - استخراج البيانات
Pre-requisites	BS214: Probability and Statistics
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduces basic data mining concepts and techniques for discovering interesting patterns hidden in large-scale data sets, focusing on issues relating to effectiveness and efficiency. Topics covered include data preprocessing, data warehouse, association, classification, clustering, and mining specific data types such as time-series, social networks, multimedia, and Web data.

IS433	Mining Social Media - تعدين وسائل الاعلام الاجتماعية
Pre-requisites	IS412: Business Intelligence
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Introduction to data mining and social media, mining Twitter; Exploring trending topics, mining Facebook; Analyzing fan pages and examining friendships, mining Instagram, mining LinkedIn, Mining Text files, Mining Web pages, Mining Mailboxes, Mining GitHub.

IS434	Advanced Database Management Systems - إدارة نظم قواعد البيانات متقدم
Pre-requisites	CF229: Database Management Systems
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	This course covers advanced aspects of database management including normalization and denormalization, query optimization, distributed databases, data warehousing, and big data. There is extensive coverage and hands on work with SQL, and database instance tuning. Course covers various modern database architectures including relational, key value, object relational and document store models as well as various approaches to scale out, integrate and implement database systems through replication and cloud based instances. Students learn about unstructured "big data" architectures and databases, and gain hands-on experience with Spark and MongoDB. Students complete a term project exploring an advanced database technology of their choice.

IS435	Multimedia Information System - نظام معلومات الوسائط المتعددة
Pre-requisites	CF216: Data Structures
Credit Hours	3
Lecture, Tutorial, Lab	2, 0, 2
Description	Multimedia data type and format, Major characteristics and requirement for multimedia data and applications, Understanding multimedia compression, Multimedia database design, Text document indexing and retrieval, Audio indexing and retrieval, Image indexing and retrieval, Video indexing and retrieval, integrated multimedia indexing and retrieval.

UR111	Human Rights - حقوق الإنسان
Pre-requisites	None
Credit Hours	1
Lecture, Tutorial, Lab	1, 0, 0
Description	(It can be offered in Arabic) Philosophical and political bases for the international human rights movement, probing the ongoing debate over universality, culture and human rights. Overview of the 20th century's benchmarks in the worldwide human rights movement. Understanding the principles outlined in the Universal Declaration of Human Rights. Types of human rights. Human rights standards. Understanding social and legal accountability. various causes of repression. Humanitarian intervention and human rights. Strategies designed to advance human rights, including: the use of military force to promote human rights; the development of international criminal courts, truth commissions, and other attempts at transitional justice; the intersection of humanitarian and human rights law, with an introduction to war crimes, crimes against humanity and the specific crime of genocide; advocacy strategies concerning globalization and transnational corporations; and the human rights intentions of terrorism.

UR112	Introduction to Computers - مقدمة في الحاسبات
Pre-requisites	None
Credit Hours	3
Lecture, Tutorial, Lab	2, 1, 1
Description	Computer types – Digital logic - Introduction to computer architecture; computer hardware, input/output units, CPU, motherboards, memory types – Storage media; Magnetic, Optical, Flash, Hybrid - Computers and networking - Information systems- Operating systems - programming languages and database concepts, Internet and Web Evolution and applications – Computer security - Ethics and health related issues.

UR113	Entrepreneurship and Innovation - ريادة الأعمال والإبتكار
Pre-requisites	None
Credit Hours	1
Lecture, Tutorial, Lab	1, 0, 0
Description	Basic concepts in the area of entrepreneurship, Role and importance of entrepreneurship for economic development, developing personal creativity and entrepreneurial initiative, adopting of the key steps in the elaboration of business idea, understanding the stages of the entrepreneurial process and the resources needed for the successful development of entrepreneurial ventures, Process of creating and managing innovative organizations. Methods of analyzing new venture opportunities and ideas; approaches to assessing and acquiring resources; strategies for acquiring existing businesses. Introduction to group psychology of organizations, organizational structure, Individuals and group dynamics, motivation, leadership, morale, power, labor-management behavior, theoretical developments in the discipline of innovation, its role in entrepreneurship and in the development of successful, cotemporary organizations.

UR122	Marketing & Digital Strategy - التسويق والإستراتيجية الرقمية
Pre-requisites	None
Credit Hours	2
Lecture, Tutorial, Lab	2, 0, 0
Description	Traditional and digital strategic approach to marketing planning, integrating digital strategy, models, frameworks and techniques throughout to achieve competitive advantage, Situation analysis, techniques to assess external and internal environments, effective decision making, Resources management, monitoring and measurement techniques.

UR211	Communication Skills - مهارات الاتصال
Pre-requisites	None
Credit Hours	2
Lecture, Tutorial, Lab	2, 0, 0
Description	Foundations of effective Inter-personal communication skills, with particular emphasis on the skills important to establishing and maintaining effective professional and personal relationships. This course provides an understanding of the fundamental and practices of oral and written skills used regularly in effective communication as well as body language, phone communication, meetings, and public presentation.

UR221	Creative & Scientific Thinking - التفكير الإبداعي والعلمي
Pre-requisites	None
Credit Hours	2
Lecture, Tutorial, Lab	2, 0, 0
Description	Intelligence and thinking, creativity, critical thinking, brainstorming, solving problems, decision making, innovation, applying the innovation, early forms of scientific methods, modern methodology, new forms of methodology, scientific methods of science and technology.

UR222	Natural Resources & Sustainability - الموارد الطبيعية والاستدامة
Pre-requisites	None
Credit Hours	1
Lecture, Tutorial, Lab	1, 0, 0
Description	Concepts and principles related to the economic, environmental, social, cultural and ethical considerations of resource management. Global natural resource issues and international collaborative efforts to address them, through the lens of sustainable development. Key drivers of natural resource policy and key stressors of natural resources and ecosystems (including population, pollution, invasive species, habitat loss and degradation, overharvest, disease, poverty, political conflict, cultural traditions, urbanization, technology and climate change), Role of ethics, social justice and communication in the sustainable management of natural resources.

UR223	Research Methods and Ethics - طرق وأخلاقيات البحث العلمي
Pre-requisites	None
Credit Hours	2
Lecture, Tutorial, Lab	2, 0, 0
Description	Scientific methods in research related to the expert area, Assessment and values of qualitative and quantitative research (induction, deduction, validity, reliability, reliability, transferability, reliability). Research designs within quantitative (experimental and not experimental research design) and qualitative research and systematic literature studies. Good academic custom and reference management at scientific writing search-ethical principles. Designs and methods for selection (strategic, randomized) data collection (questionnaire, interviews, and observations) and analysis (descriptive and analytical statistics; describing and interpreting qualitative analysis) related to AI and issues.